Past Problems

for (MATH200) Differential Equations

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POSTECH

Problem 1. (2012 Spring POSTECH Differential Equations). In the theory of memory, we assume that the rate at which a subject is memorized is proportional to the amount that is left to be memorized. Suppose that M is the total amount of a subject to be memorized and $A(t)$ is the amount memorized in time t. Write a differential equation for the amount $A(t)$ and solve that equation.

Problem 2. (2012 Spring POSTECH Differential Equations). Find the general solutions of the following ordinary differential equations.

(a) $y' = \dfrac{x-2y}{3x-6y+4}$.

(b) $y' = \dfrac{3x-y-9}{x+y+1}$.

(c) $L[y] = x^2y'' - xy' + y = \ln x$, $x > 0$. (For the solution of $L[y] = 0$, guess solutions of the form $y(x) = x^r$.)

(d) $(2y^2 + 3x)\,\mathrm{d}x + (2xy)\,\mathrm{d}y = 0$.

Problem 3. (2012 Spring POSTECH Differential Equations). Find two solutions of Bessel's equation

$$x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0, \qquad 0 < x < \infty,$$

using the series solution method. Derive the recurrence relation and use it to find two solutions.

Problem 4. (2012 Spring POSTECH Differential Equations). Solve the initial-value problem

$$3t^2y + 8ty^2 + (t^3 + 8t^2y + 12y^2)\frac{\mathrm{d}y}{\mathrm{d}t} = 0, \qquad y(2) = 1.$$

Problem 5. (2012 Spring POSTECH Differential Equations). Consider the differential equation

$$y'' + ty' + y = 0 \tag{1}$$

on the interval $-\infty < t < \infty$.

(a) Show that one solution is $y_1(t) = e^{-t^2/2}$ and use reduction of order to show that a second solution is

$$y_2(t) = e^{-t^2/2}\int_0^t e^{s^2/2}\,\mathrm{d}s.$$

(b) Compute $W(y_1, y_2)(t)$.
(c) Show that y_1 and y_2 form a fundamental set of solutions of (1) on $-\infty < t < \infty$.
(d) Solve the initial-value problem $y'' + ty' + y = 0$, $y(0) = 0$, $y'(0) = 1$.

Problem 6. (2012 Spring POSTECH Differential Equations). Consider the second-order differential equation $y'' + y = 8\cos t$.

(a) Find a fundamental set of solutions of the homogeneous equation $y'' + y = 0$ (check the Wronskian at some point).
(b) Find a particular solution of the inhomogeneous equation.
(c) Assume that the particular solution found in part (b) describes the movement of a bridge. What will happen to the bridge for large t?

Problem 7. (2012 Spring POSTECH Differential Equations). Consider the initial-value problem

$$y'' - 3y' + 2y = e^{3t}, \qquad y(0) = 1, \quad y'(0) = 0.$$

(a) Solve the equation without using the Laplace transform.
(b) Solve the equation using the Laplace transform.

Problem 8. (2012 Spring POSTECH Differential Equations).

(a) Solve the boundary-value problem for $0 < t < 1$:

$$y'' + \frac{1}{t}y' = -1, \qquad y(1) = 0,$$

with $y(t)$ bounded for $0 < t < 1$.
(b) Determine the condition on initial values so that all solutions of

$$\mathbf{x}' = \begin{pmatrix} -1 & 6 \\ 1 & -2 \end{pmatrix} \mathbf{x}$$

are bounded for all $t \geq 0$.

Problem 9. (2012 Spring POSTECH Differential Equations). Solve

$$\mathbf{x}' = \begin{pmatrix} 7 & -1 \\ 1 & 5 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2e^{6t} \\ 6te^{6t} \end{pmatrix}.$$

Problem 10. (2012 Spring POSTECH Differential Equations). Find the Fourier-series solution $y(t)$ satisfying

$$y'' + 4y = f(t), \qquad y(0) = 1, \quad y'(0) = 0,$$

where f is periodic with period 2 and

$$f(t) = \begin{cases} -t + 1, & 0 \leq t < 1, \\ t - 1, & 1 \leq t < 2. \end{cases}$$

Problem 11. (2012 Spring POSTECH Differential Equations). Solve the heat equation

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}, \qquad 0 < x < L, \quad t > 0, \quad k > 0,$$

where

$$u(x,0) = \sin\frac{\pi x}{L} \quad (0 \le x \le L), \qquad u(0,t) = u(L,t) = 0 \quad (t \ge 0),$$

and describe how the temperature changes as time progresses.

Problem 12. (2012 Spring POSTECH Differential Equations). Consider the two-dimensional wave equation

$$c^2(u_{xx} + u_{yy}) = u_{tt}, \qquad 0 < x < a, \quad 0 < y < b, \quad t > 0,$$

with

$$u(x,y,0) = f(x,y), \qquad u_t(x,y,0) = g(x,y), \quad 0 \le x \le a, \quad 0 \le y \le b,$$

and

$$u(0,y,t) = u(a,y,t) = u(x,0,t) = u(x,b,t) = 0$$

for $0 \le x \le a$, $0 \le y \le b$, $t \ge 0$. By separation of variables, put $u(x,y,t) = X(x)Y(y)T(t)$.

(a) Find three ordinary differential equations satisfied by X, Y, and T. *(Hint. You may start with $u(x,y,t) = F(x,y)T(t)$.)*

(b) Express the conditions in terms of eigenvalues and find the boundary conditions of the differential equations satisfied by X, Y for nontrivial solutions.

(c) *(Bonus)* Express the solution $u(x,y,t)$.

Problem 13. (2012 Spring POSTECH Differential Equations). Find the solution $u(r,\theta)$ of Laplace's equation

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$$

outside the circle $r = a$, satisfying

$$u(a,\theta) = f(\theta), \qquad 0 \le \theta \le 2\pi.$$

Problem 14. (2013 Spring POSTECH Differential Equations). Solve

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x+y+2}{x-y+4}, \qquad y(-2) = 1.$$

(Hint. Let $x = X - a$, $y = Y - b$, and choose a, b suitably to remove the constants 2 and 4.)

Problem 15. (2013 Spring POSTECH Differential Equations). Find an integrating factor and solve

$$(y^2 + 4ye^x) + 2(y + e^x)\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Problem 16. (2013 Spring POSTECH Differential Equations). A tank originally contains 12 liters of fresh water. Water containing 2 g of salt per liter is poured into the tank at a rate of 3 liters per minute, and the well-stirred mixture is allowed to leave at the same rate. After 8 minutes the process is stopped, and fresh water is poured into the tank at a rate of 2 liters per minute, with the mixture leaving at the same rate. Find the amount of salt in the tank at t minutes after the first 8 minutes (that is, at $t + 8$ minutes from the beginning).

Problem 17. (2013 Spring POSTECH Differential Equations). Find a second solution of

$$x^2y'' + (x^2 - 2x)y' - (x - 2)y = 0,$$

given $y_1(x) = x$.

Problem 18. (2013 Spring POSTECH Differential Equations). Find a general solution using the method of undetermined coefficients:

$$y'' + 4y' + 4y = 7t + 6te^{-2t}, \qquad y(0) = 0, \quad y'(0) = 0.$$

Problem 19. (2013 Spring POSTECH Differential Equations). Using variation of parameters, find a general solution of

$$y'' - 3y' + 2y = \sin(e^{-t}).$$

Problem 20. (2013 Spring POSTECH Differential Equations). Answer the following questions.

(a) Let $y(x) = \sum_{n=0}^{\infty} a_n x^n$ be a series solution of

$$y'' - x^2y' - xy = 0.$$

Find two solutions $y_1(x)$ and $y_2(x)$ such that $y(x) = a_0y_1(x) + a_1y_2(x)$.

(b) Show that y_1 and y_2 form a fundamental set of solutions for all x.

Problem 21. (2013 Spring POSTECH Differential Equations). Determine the lower bound for the radius of convergence of the series solutions of

$$(1 + x^2)y'' + y' + (\sin x)(1 + x^2)y = 0$$

about $x = 0$.

Problem 22. (2013 Spring POSTECH Differential Equations). Answer the following questions.

(a) Show that

$$4x^2y'' + 8\ln(1 + x)y' + y = 0$$

has a regular singular point at $x = 0$. *(Hint. Note that* $\ln(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$, $-1 < x \leq 1$.*)*

(b) Let $y_1(x) = \sum_{n=0}^{\infty} a_n x^{r+n}$ be a solution ($x > 0$, $a_0 \neq 0$). Determine r.

(c) Show that the second solution is of the form

$$y_2(x) = y_1(x)\ln x + x^r \sum_{n=1}^{\infty} b_n x^n \qquad (x > 0),$$

where $y_1(x)$ and r are given in part (b).

Problem 23. (2013 Spring POSTECH Differential Equations). Use the Laplace transform to solve the following problems.

(a) $y'' + 9y = \cos(\omega t)$, $y(0) = 0$, $y'(0) = 1$, where $\omega \neq 3$ and $\omega > 0$.

(b) $y'' + 9y = \cos(3t)$, $y(0) = 0$, $y'(0) = 1$.

(c) In which of the above two cases is it possible for the magnitude (or absolute value) of a solution to grow arbitrarily large? Explain mathematically.

Problem 24. (2013 Spring POSTECH Differential Equations). Find the inverse Laplace transform of

$$F(s) = \frac{1}{(s^2+9)^2}.$$

(Hint. Consider the derivative of $s/(s^2+9)$.)

Problem 25. (2013 Spring POSTECH Differential Equations). Compute the Laplace transform of

$$f(t) = \begin{cases} t^2, & 0 \le t < 2, \\ 4t, & 2 \le t < \infty \end{cases}$$

using a unit-step function.

Problem 26. (2013 Spring POSTECH Differential Equations). Find the solution of the initial-value problem

$$y'' - y = \delta(t-1), \qquad y(0) = 1, \quad y'(0) = 0.$$

Problem 27. (2013 Spring POSTECH Differential Equations). Find the solution, in terms of a convolution integral, of the initial-value problem

$$y'' + 9y = g(t), \qquad y(0) = 0, \quad y'(0) = 1.$$

Problem 28. (2013 Spring POSTECH Differential Equations). Solve the system

$$t\mathbf{x}' = \begin{pmatrix} 6 & 2 \\ 4 & 4 \end{pmatrix} \mathbf{x}.$$

Problem 29. (2013 Spring POSTECH Differential Equations). Find two real-valued independent solutions (and show their independence) of the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{pmatrix} 2 & -4 \\ 1 & 2 \end{pmatrix}.$$

Problem 30. (2013 Spring POSTECH Differential Equations). Consider the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Solve the system and find the fundamental matrix $\mathbf{\Phi}(t)$ such that $\mathbf{\Phi}(0) = \mathbf{I}$ (the identity matrix).

(b) Write the solution $\mathbf{x}(t)$ satisfying

$$\mathbf{x}(0) = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}.$$

Problem 31. (2013 Spring POSTECH Differential Equations). Two tanks are interconnected. Tank 1 initially contains 2 L of water with 150 g of salt, and Tank 2 initially contains 1 L of water with 50 g of salt. At $t = 0$, pure water begins to flow into Tank 1 at 3 L/min, and well-mixed salt water is exchanged from Tank 1 to Tank 2 at 4 L/min and from Tank 2 to Tank 1 at 3 L/min. Salt water is drained at 2 L/min from Tank 1 and at 1 L/min from Tank 2.

(a) Formulate the system of differential equations for the quantities of salt $x_1(t)$ and $x_2(t)$ in Tanks 1 and 2 at time $t > 0$.

(b) Solve the system, finding $x_1(t)$ and $x_2(t)$ and their limits as $t \to \infty$.

Problem 32. (2013 Spring POSTECH Differential Equations). Assume that all eigenvalues are real. Find the eigenvalues and eigenfunctions of the following boundary-value problem ($x > 0$):

$$x^2y'' + xy' - \lambda y = 0, \qquad y(1) = 0, \quad y(3) = 0.$$

Problem 33. (2013 Spring POSTECH Differential Equations). Suppose that $f(x) = 2 - 2x$ for $0 \le x \le 1$.

(a) Find the Fourier series of the even periodic extension of f with period 2.

(b) Briefly explain why the Fourier series in part (a) converges to the even periodic extension of f for all real x.

(c) Show that

$$\frac{\pi^2}{8} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}.$$

Problem 34. (2013 Spring POSTECH Differential Equations). Consider a metal rod of length 10, for which $\alpha^2 = 1$. Suppose the initial temperature distribution is $f(x) = x$, and the boundary conditions are $u(0, t) = 20$ and $u(10, t) = 0$ for $t > 0$. Find the temperature $u(x, t)$ in the rod.

Problem 35. (2013 Spring POSTECH Differential Equations). Answer the following questions.

(a) Show that the eigenvalues of the following boundary-value problem are real:

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(L) = 0 \quad (L \neq 0).$$

(b) Consider an elastic string of length 2π, whose ends are held fixed. The string is set in motion from its equilibrium position with initial velocity $u_t(x, 0) = f(x)$. Take $a = 1$ and $f(x) = 1$ if $\pi - 1 < x < \pi + 1$, and $f(x) = 0$ otherwise. Find the displacement $u(x, t)$.

Problem 36. (2013 Spring POSTECH Differential Equations). Find the solution $u(x, y)$ of Laplace's equation in the rectangle $0 < x < 2$, $0 < y < 1$, satisfying the boundary conditions

$$u(0, y) = u(2, y) = 0 \quad (0 < y < 1), \qquad u(x, 0) = 0, \quad u(x, 1) = f(x) \quad (0 < x < 2),$$

where

$$f(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 < x \le 2. \end{cases}$$

Problem 37. (2013 Fall POSTECH Differential Equations). Let $y_1(t) = t^2 + 1$, $y_2(t) = t^2 - 1$ on $I = (-2, 1)$.

(a) Compute $W(y_1, y_2)(t)$.

(b) Are $y_1(t)$ and $y_2(t)$ linearly independent on I? Why?

(c) Can $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of an equation

$$y'' + p(t)y' + q(t)y = 0 \qquad \text{on } I,$$

with $p(t)$ and $q(t)$ continuous on I? Justify with complete details.

Problem 38. (2013 Fall POSTECH Differential Equations). Find the general solutions of the following ordinary differential equations.

(a) $y''' - y'' + y' - y = \sin x$.

(b) $\dfrac{\mathrm{d}^4 y}{\mathrm{d}t^4} + 2\dfrac{\mathrm{d}^2 y}{\mathrm{d}t^2} + y = 0$.

Problem 39. (2013 Fall POSTECH Differential Equations). Solve the initial-value problems and indicate the maximal intervals of existence.

(a) $y' + y = e^{-t}t^{-1}$, with $y(2) = 1$.

(b) $(2x - y)\,\mathrm{d}x + (2y - x)\,\mathrm{d}y = 0$, with $y(1) = 3$.

Problem 40. (2013 Fall POSTECH Differential Equations). Consider the third-order linear ordinary differential equation

$$y''' + p(t)y'' + q(t)y' + r(t)y = 0 \tag{$*$}$$

on $I = (-1, 1)$, where $p(t)$, $q(t)$, and $r(t)$ are continuous on I.

(a) State Abel's theorem for equation $(*)$.

(b) Prove the stated Abel's theorem.

(c) Can $y = t^4$ be a solution of $(*)$ satisfying $y(0) = y'(0) = y''(0) = 0$? Justify with complete details.

Problem 41. (2013 Fall POSTECH Differential Equations). Find all solutions of

$$\left(y'' + \frac{3}{x}y' + \frac{1}{x^2}y\right)\left(y'' + \frac{3}{2x}y' - \frac{1}{2x^2}y\right) = 0, \qquad x \in (0, \infty).$$

Problem 42. (2013 Fall POSTECH Differential Equations). On $I = (0, 2)$, solve

$$y'' + (x-1)^2 y' + (x^2 - 1)y = 0, \qquad y(1) = 0, \quad y'(1) = 1.$$

Problem 43. (2013 Fall POSTECH Differential Equations). Determine whether each statement is true or false.

(1) $y'' + e^{\sin x}y' + y^2 = x^4$ is a linear ordinary differential equation.

(2) In the population model $y' = r(y - T)(K - y)y$, with $r > 0$ and $K > T > 0$, $y = T$ is an unstable equilibrium solution.

(3) If f and $\frac{\partial f}{\partial t}$ are bounded on I and $t_0 \in I$, then $y' = f(t, y)$ with $y(t_0) = 1$ has a unique solution on I.

(4) $x = 0$ is not an irregular singular point for $(x + 1)^2 y'' + 6y = 0$.

(5) We can determine a series solution about $x = 0$ for $y'' + (\sin x)y' + (x + x^3)y = 0$.

Problem 44. (2014 Spring POSTECH Differential Equations). Solve the initial-value problem

$$4t^3e^{t+y} + t^4e^{t+y} + 2t + \left(t^4e^{t+y} + 2y\right)\frac{\mathrm{d}y}{\mathrm{d}t} = 0, \qquad y(0) = 1.$$

Problem 45. (2014 Spring POSTECH Differential Equations). Consider the ordinary differential equation

$$2t^2y'' + 3ty' - y = 0, \qquad 0 < t < \infty.$$

(a) Show that $y_1(t) = \sqrt{t}$ and $y_2(t) = 1/t$ are solutions.
(b) Compute $W(y_1, y_2)(t)$. What happens as t approaches zero?
(c) Show that y_1 and y_2 form a fundamental set of solutions on $0 < t < \infty$.
(d) Solve the equation with $y(1) = 2$ and $y'(1) = 1$.

Problem 46. (2014 Spring POSTECH Differential Equations). Solve the following problems.

(a) Find the general solution of $y'' - y = 0$.
(b) Use variation of parameters to find general solutions of

$$L[y] = y'' - 5y' + 6y = 4e^t.$$

(c) Solve

$$y'' + 4y = 3\sin(2t), \qquad y(0) = 0, \quad y'(0) = 0.$$

Problem 47. (2014 Spring POSTECH Differential Equations). Consider

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 5t\frac{\mathrm{d}y}{\mathrm{d}t} + 3y = 0. \tag{1}$$

(a) Determine the type of the point $t = 0$.
(b) Find the solution of (1) subject to $y(1) = 1$ and $y'(1) = 5$.
(c) Rewrite (1) using the change of variable $x = \ln t$ on $0 < t < \infty$.
(d) Find the general solution of (1) on $0 < t < \infty$.

Problem 48. (2014 Spring POSTECH Differential Equations). Using the table of Laplace transforms, solve the initial-value problem

$$y' - t * y = \delta(t - 1), \qquad y(0) = 0,$$

where $*$ denotes convolution.

Problem 49. (2014 Spring POSTECH Differential Equations). Find a power-series solution about $x = 1$ of

$$y'' + (x-1)^2y' + ((x-1)^2 + 2(x-1))y = 0,$$

with $y(1) = 1$, $y'(1) = 0$.

Problem 50. (2014 Spring POSTECH Differential Equations). Consider the following initial-value problem:

$$y'' - 3y' + 2y = e^{3t}, \qquad y(0) = 1, \quad y'(0) = 0.$$

(a) Solve the equation without using the Laplace transform.
(b) Solve the equation using the Laplace transform.

Problem 51. (2014 Spring POSTECH Differential Equations). Let

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & \alpha \end{pmatrix},$$

where α is a constant. Let $W(t)$ be a Wronskian for

$$\mathbf{x}' = \mathbf{A}\mathbf{x}. \tag{1}$$

Given that $W(t)$ satisfies $W'(t) + 2W(t) = 0$,

(a) Find $e^{\mathbf{A}t}$.

(b) Solve (1) with initial condition

$$\mathbf{x}(0) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Problem 52. (2014 Spring POSTECH Differential Equations). Let $u(x,t)$ be the temperature in a uniform rod of length L and thermal diffusivity $\alpha^2 = \frac{\kappa}{\rho s}$. Given initial temperature

$$u(x,0) = \sin\frac{\pi x}{L}, \qquad 0 \le x \le L,$$

assume both ends of the rod are insulated:

$$u_x(0,t) = u_x(L,t) = 0 \qquad (t > 0).$$

(a) Find the temperature $u(x,t)$.

(b) What is the steady-state temperature as $t \to \infty$?

(c) Briefly describe how the temperature changes as time progresses.

Problem 53. (2014 Spring POSTECH Differential Equations). Find the solution $u(r,\theta)$ of Laplace's equation in polar coordinates

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, \qquad r > a,$$

outside the circle $r = a$ for a constant $a > 0$, satisfying

$$u(a,\theta) = f(\theta), \qquad 0 \le \theta < 2\pi,$$

and the asymptotic behavior

$$u(r,\theta) \to 0 \qquad (r \to +\infty).$$

Problem 54. (2014 Spring POSTECH Differential Equations). Discuss solutions of the initial-boundary-value problem

$$\begin{cases} u_t = u_{xx} - x, & x \in (0,1),\ t > 0, \\ u(0,t) = 0, & t > 0, \\ u_x(1,t) = 0, & t > 0, \\ u(x,0) = \sin(\pi x/2), & x \in [0,1]. \end{cases}$$

Problem 55. (2014 Spring POSTECH Differential Equations). Consider the initial-boundary-value problem for the wave equation

$$\begin{cases} u_{tt} - u_{xx} = 0, & 0 < x < 1,\ t > 0, \\ u_x(0,t) = 0, \quad u(1,t) = 1, & t > 0, \\ u(x,0) = \cos(2\pi x), & 0 \le x \le 1, \\ u_t(x,0) = 0, & 0 \le x \le 1. \end{cases} \tag{2}$$

(a) Find a solution of (2).

(b) Prove that the solution is unique. *(Hint. Assume that $u_1(x,t)$ and $u_2(x,t)$ are two solutions of (2), and try to show that $v = u_1 - u_2 \equiv 0$.)*

Problem 56. (2014 Spring POSTECH Differential Equations). Consider the eigenvalue problem

$$\begin{cases} -y'' = \lambda y, & x \in (0,1), \\ y(0) - 3y'(0) = 0, \quad y'(1) = 0. \end{cases} \tag{3}$$

(a) Find the eigenvalues of (3).

(b) Determine the corresponding normalized eigenfunctions. *(Hint. A function ϕ is normalized if $\int_0^1 \phi^2(x)\,\mathrm{d}x = 1$.)*

Problem 57. (2014 Fall POSTECH Differential Equations).

(a) Solve the Bernoulli equation

$$t^2 y' + ty = y^4, \qquad y(1) = 2.$$

Give y explicitly as a function of t. *(Hint. Let $y = v^a$ and choose a to obtain a linear equation.)*

(b) State the largest interval on which the solution is valid, and briefly explain why no other solution exists there.

Problem 58. (2014 Fall POSTECH Differential Equations). Solve

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x + y + 1}{x - y + 3}, \qquad x > -2.$$

(Hint. Let $x = X + a$, $y = Y + b$, and choose a, b to remove the constant terms.)

Problem 59. (2014 Fall POSTECH Differential Equations). Find an integrating factor to solve

$$(y^2 + 3ye^x) + (y + e^x)\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Problem 60. (2014 Fall POSTECH Differential Equations). Write an integral equation equivalent to the initial-value problem

$$y' = t^2 + y^2, \qquad y(0) = 1.$$

Then find the successive approximations $\phi_1(t)$ and $\phi_2(t)$ starting from $\phi_0(t) = 0$.

Problem 61. (2014 Fall POSTECH Differential Equations). Consider the differential equation

$$y'' - 3ty' + 3y = 0, \qquad t > 0.$$

(a) First find one solution by guessing.

(b) Use reduction of order to find the general solution. Your solution may involve a definite integral beginning at $t_0 = 1$.

(c) Use the Wronskian to verify that there are no other solutions.

Problem 62. (2014 Fall POSTECH Differential Equations). Use the method of undetermined coefficients to find the general solution of

$$y'' - 4y' + 4y = t + e^{2t}.$$

Problem 63. (2014 Fall POSTECH Differential Equations). A mass weighing 16 lb stretches a spring 2 ft (assume $g = 32$ ft/s^2). The mass is in a medium that exerts a viscous resistance of 4 lb when it has a velocity of 2 ft/s. If the mass is set in motion from its equilibrium position with downward velocity of 3 ft/s, find its position y at time t.

Problem 64. (2014 Fall POSTECH Differential Equations).

(a) Consider $y'' + p(t)y' + q(t)y = g(t)$. Assuming $y = c_1y_1 + c_2y_2$ is the general solution of the homogeneous equation, derive an expression for a particular solution using variation of parameters.

(b) Find a particular solution of

$$y'' + y = \tan t, \qquad 0 < t < \frac{\pi}{2}.$$

(Recall that $\frac{\mathrm{d}}{\mathrm{d}t} \ln|\sec t + \tan t| = \sec t$*.)*

Problem 65. (2014 Fall POSTECH Differential Equations). Show that

$$2x^2y'' - xy' + (x+1)y = 0$$

has a regular singular point at $x = 0$, and find a series solution corresponding to the larger root of the indicial equation (the characteristic equation of the corresponding Euler equation). Find a simplified expression for the nth term in the series.

Problem 66. (2014 Fall POSTECH Differential Equations). Use the Laplace transform to solve

$$y'' + 9y = \cos(3t), \qquad y(0) = 0, \quad y'(0) = 1.$$

Problem 67. (2014 Fall POSTECH Differential Equations). Let $f(t) = t$ for $t < 3$ and $f(t) = 0$ for $t \geq 3$. Use the Laplace transform to solve

$$y'' + 4y = f(t), \qquad y(0) = 0, \quad y'(0) = 0.$$

Problem 68. (2014 Fall POSTECH Differential Equations). Solve the initial-value problem

$$y'' - 2y = \delta(t-1), \qquad y(0) = 1, \quad y'(0) = 1.$$

Problem 69. (2014 Fall POSTECH Differential Equations). Use the method of undetermined coefficients to find the general solution of

$$y'' - 9y' + 9y = t^2 + e^{3t}.$$

Problem 70. (2014 Fall POSTECH Differential Equations). A tank originally contains 8 liters of fresh water. Water containing 2 g of salt per liter is poured into the tank at a rate of 3 liters per minute, and the well-stirred mixture leaves at the same rate. After 12 minutes the process is stopped, and fresh water is poured into the tank at a rate of 2 liters per minute, with the mixture leaving at the same rate. Find the amount of salt in the tank at t minutes after the first 12 minutes (that is, at $t + 12$ minutes from the beginning).

Problem 71. (2014 Fall POSTECH Differential Equations). Find the general solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$, where $\mathbf{x} = \mathbf{x}(t)$ and

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix}.$$

Problem 72. (2014 Fall POSTECH Differential Equations).

(a) Find a fundamental matrix $\mathbf{\Phi}(t)$ with $\mathbf{\Phi}(0) = \mathbf{I}$ for

$$\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 1 & -2 \end{pmatrix} \mathbf{x}, \qquad \mathbf{x} = \mathbf{x}(t).$$

(b) Use the result of part (a) to find the solution satisfying

$$\mathbf{x}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Problem 73. (2014 Fall POSTECH Differential Equations). Find the solution of $\mathbf{x}' = \mathbf{A}\mathbf{x}$, where $\mathbf{x} = \mathbf{x}(t)$,

$$\mathbf{A} = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix}, \qquad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Problem 74. (2014 Fall POSTECH Differential Equations). Assume that the 2π-periodic function $f(x)$ has the Fourier series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

(a) Formally (assuming that integration and summation can be interchanged), show Parseval's equation:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2 \, \mathrm{d}x = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

(b) Find the Fourier series of $f(x) = x$, $-\pi \le x < \pi$, extended to be 2π-periodic.

(c) Show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Problem 75. (2014 Fall POSTECH Differential Equations). Derive the heat equation $u_{xx}(x,t) = u_t(x,t)$. Consider a straight metal rod of length π, placed so that $0 < x < \pi$, with uniform cross-sectional area A and insulated sides. Assume thermal conductivity $\kappa = 1$, specific heat $s = 1$, and density $\rho = 1$, so that $\alpha^2 = \kappa/(\rho s) = 1$. Show that the temperature $u(x,t)$ satisfies $u_{xx}(x,t) = u_t(x,t)$ for $0 < x < \pi$, $t > 0$.

Problem 76. (2014 Fall POSTECH Differential Equations). Consider a metal rod of length 2, with $\alpha^2 = 1$. Suppose the initial temperature distribution is $f(x) = x^2$ on $0 < x < 2$, and the boundary conditions are $u(0,t) = 6$ and $u(2,t) = 0$ for $t > 0$. Find the temperature $u(x,t)$ in the rod.

Problem 77. (2014 Fall POSTECH Differential Equations). Consider an elastic string of length 4π, whose ends are held fixed. The string is set in motion from its equilibrium position with initial velocity $u_t(x,0) = g(x)$. Take $a = 1$, and let $g(x) = 2$ if $\pi < x < 3\pi$, and $g(x) = 0$ for $0 < x < \pi$ and $3\pi < x < 4\pi$. Find the displacement $u(x,t)$.

Problem 78. (2014 Fall POSTECH Differential Equations). Find the solution $u(x,y)$ of Laplace's equation in the rectangle $0 < x < 4$, $0 < y < 1$, satisfying

$$u(0,y) = u(4,y) = 0 \quad (0 < y < 1), \qquad u(x,0) = 0, \quad u(x,1) = f(x) \quad (0 < x < 4),$$

where

$$f(x) = \begin{cases} x, & 0 \le x \le 2, \\ 4 - x, & 2 < x \le 4. \end{cases}$$

Problem 79. (2015 Spring POSTECH Differential Equations). Consider the initial-value problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t,y), \qquad y(t_0) = y_0.$$

(a) State the conditions for existence and uniqueness of solutions to the initial-value problem.

(b) Set $f(t,y) = (y-1)^{2/3}$ for $t \ge 0$ and $(t_0, y_0) = (0,1)$. Find the solution to the initial-value problem. What can you say about uniqueness of the solution?

Problem 80. (2015 Spring POSTECH Differential Equations). Solve the following equations.

(a) $(y\cos x + 2xe^y) + (\sin x + x^2e^y - 1)y' = 0$, $y(0) = 1$.

(b) $3xy + y^2 + (x^2 + xy)y' = 0$.

Problem 81. (2015 Spring POSTECH Differential Equations). Solve the following equations.

(a) $xy' = \dfrac{y^2}{x} + y$, $y(1) = -1$.

(b) $y' + \dfrac{1}{x}y = 3x^2y^3$, $y(1) = 1$.

Problem 82. (2015 Spring POSTECH Differential Equations). Given that $y_1(x) = x$ is a solution of

$$(x-1)y'' - xy' + y = 0, \qquad x > 1,$$

find a fundamental set of solutions.

Problem 83. (2015 Spring POSTECH Differential Equations). Solve the initial-value problem

$$y'' + y' + y = f(t), \qquad y(0) = 3, \quad y'(0) = 5,$$

without using the Laplace transform.

Problem 84. (2015 Spring POSTECH Differential Equations). Solve the nonhomogeneous Bessel equation

$$x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = x\sqrt{x}, \qquad x > 0, \quad y(\pi) = 0, \quad y'(\pi) = 1.$$

Hint. Use the substitution $y = x^{-1/2}v$.

Problem 85. (2015 Spring POSTECH Differential Equations). Consider the second-order differential equation

$$y'' + p(x)y' + q(x)y = 0,$$

where $p(x) = \dfrac{x^2+1}{x-3}$ and $q(x) = (x-3)^{-2}$. Find the regular singular point x_0 and the indicial equation, its roots, the series solution, and the radius of convergence.

Problem 86. (2015 Spring POSTECH Differential Equations). Consider the initial-value problem

$$y'' + y' + y = \delta(t-5), \qquad y(0) = y'(0) = 0,$$

where δ is the Dirac delta function. Find the solution.

Problem 87. (2015 Spring POSTECH Differential Equations). Find all y satisfying the following conditions for some $\lambda \in \mathbb{R}$:

$$y'' + \lambda y = 0, \qquad y(0) - y'(0) = 0, \quad y(1) = 0.$$

Problem 88. (2015 Spring POSTECH Differential Equations). Find all $y(x)$ such that, for some constant $\lambda \in \mathbb{R}$,

$$x^2y'' + xy' + \lambda y = 0, \qquad \lim_{x\to 0} y(x) \text{ exists and is finite.}$$

Problem 89. (2015 Spring POSTECH Differential Equations). Determine a fundamental matrix for

$$\mathbf{A} = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix}.$$

Problem 90. (2015 Spring POSTECH Differential Equations). Determine the solution to

$$\mathbf{x}'(t) = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix}\mathbf{x}(t) + \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}, \qquad \mathbf{x}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Problem 91. (2015 Spring POSTECH Differential Equations). Let $\mathbf{A}$ be a real $n \times n$ matrix and $\mathbf{A}^\mathsf{T}$ its transpose. Denote the inner product by $\langle \mathbf{x}, \mathbf{y} \rangle$. Show that if $\det \mathbf{A} = 0$ and $\mathbf{Ax} = \mathbf{b}$ has a solution, then $\langle \mathbf{b}, \mathbf{y} \rangle = 0$ for every solution $\mathbf{y}$ of $\mathbf{A}^\mathsf{T}\mathbf{y} = \mathbf{0}$.

Problem 92. (2015 Spring POSTECH Differential Equations). Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 2 & -1 & 3 \end{pmatrix}.$$

Compute $\det \mathbf{A}$. Find the condition on $\mathbf{b}$ such that $\mathbf{Ax} = \mathbf{b}$ has a solution, and find $\mathbf{x}$.

Problem 93. (2015 Spring POSTECH Differential Equations). Find the general solution of the system $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$, where

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -4 & -1 \end{pmatrix}.$$

Problem 94. (2015 Spring POSTECH Differential Equations). Let $f(x) = x$ for $-1 < x < 1$, with $f(-1) = f(1) = 0$. Find the Fourier series of $f(x)$ and show that

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Problem 95. (2015 Spring POSTECH Differential Equations). Find the solution of the heat equation

$$4u_t = u_{xx}, \qquad 0 < x < 2, \quad t > 0,$$

subject to

$$u(0,t) = u(2,t) = 0, \qquad u(x,0) = 2\sin\frac{\pi x}{2} - \sin(\pi x) + 4\sin(2\pi x).$$

Problem 96. (2015 Spring POSTECH Differential Equations). Let $a > 0$, and let $f(x)$ and $g(x)$ be smooth functions. Set

$$u(x,t) = \frac{1}{2}(f(x-at) + f(x+at)) + \frac{1}{2a}\int_{x-at}^{x+at} g(s)\, \mathrm{d}s.$$

Show that $u(x,t)$ satisfies the wave equation

$$a^2 u_{xx} = u_{tt}, \qquad -\infty < x < \infty, \quad t > 0,$$

with $u(x,0) = f(x)$ and $u_t(x,0) = g(x)$ for all $x \in \mathbb{R}$.

Problem 97. (2015 Spring POSTECH Differential Equations). Consider the Laplace equation

$$u_{xx} + u_{yy} = 0, \qquad 0 < x, y < 1,$$

with boundary conditions

$$u(x,0) = 0 = u(x,1), \quad 0 < x < 1,$$
$$u(0,y) = 0, \quad u(1,y) = e^y, \quad 0 \le y \le 1.$$

Solve the equation and explain why the solution series converges.

Problem 98. (2015 Spring POSTECH Differential Equations).

(a) Let $x = r\cos\theta$, $y = r\sin\theta$ be the polar-coordinate transformation. Show that $u_{xx} + u_{yy}$ becomes

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}.$$

(b) Describe the solution to

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, \qquad 0 < r < a,$$

with $u(a,\theta) = g(\theta)$, where $a > 0$ and g is continuous and 2π-periodic, and (r,θ) are the polar coordinates on $\mathbb{R}^2$.

Problem 99. (2015 Spring POSTECH Differential Equations). Consider the Sturm–Liouville boundary-value problem

$$-u''(x) + u(x) = \lambda u(x), \quad 0 < x < 1, \qquad u'(0) = 0, \quad u(1) = 0.$$

Define $L[u] = -u'' + u$. Suppose u and v both satisfy the boundary conditions above.

(a) Show that

$$\int_0^1 L[u]v \, \mathrm{d}x = \int_0^1 L[v]u \, \mathrm{d}x.$$

(b) Show that the eigenvalues λ of the problem are real.

Problem 100. (2015 Fall POSTECH Differential Equations). Solve the following differential equations.

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x+y}{x-y}$, $y(1) = 0$.

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2x-5y}{6x-15y+4}$. *Hint.* Note that $\mathrm{d}/\mathrm{d}x$ is linear, and change the equation to a convenient form.

Problem 101. (2015 Fall POSTECH Differential Equations). Find an integrating factor and solve each equation. *Hint.* Choose a proper variable t such that $\mu = \mu(t)$.

(a) $2(x + e^y) + (x^2 + 4xe^y)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0.$

(b) $\left(3x + \dfrac{6}{y}\right) + \left(\dfrac{x^2}{y} + \dfrac{3y}{x}\right)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0.$

Problem 102. (2015 Fall POSTECH Differential Equations). A tank originally contains 12 liters of fresh water. Then water containing 2 grams of salt per liter is poured into the tank at a rate of 3 liters per minute, and the well-stirred mixture is allowed to leave at the same rate. After 8 minutes the process is stopped, and fresh water is poured into the tank at a rate of 2 liters per minute, with the mixture leaving at the same rate. Find the amount of salt in the tank at t minutes after the first 8 minutes (i.e. at $t + 8$ minutes from the beginning).

Problem 103. (2015 Fall POSTECH Differential Equations). Using reduction of order, find a second solution $y_2(x)$ to

$$y'' + xy' + y = 0, \qquad y_1(x) = e^{-x^2/2},$$

and show that the two solutions are linearly independent.

Problem 104. (2015 Fall POSTECH Differential Equations). Find a solution of the initial-value problem using the method of undetermined coefficients:

$$y'' + 2y' + y = \sin t + 6te^{-t}, \qquad y(0) = 0, \quad y'(0) = 0.$$

Problem 105. (2015 Fall POSTECH Differential Equations). Determine the lower bound for the radius of convergence of the series solutions of

$$(1 + x^2)y'' + y' + (\sin x)(1 + x^2)y = 0$$

about $x = 0$.

Problem 106. (2015 Fall POSTECH Differential Equations). (Variation of parameters.) Let $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of

$$y'' + p(t)y' + q(t)y = 0.$$

(a) Prove that if $\psi(t) = u(t)y_1(t) + v(t)y_2(t)$ is a particular solution to

$$y'' + p(t)y' + q(t)y = g(t),$$

then the coefficient functions can be chosen to satisfy

$$u'(t) = -\frac{g(t)y_2(t)}{W(y_1, y_2)(t)}, \qquad v'(t) = \frac{y_1(t)g(t)}{W(y_1, y_2)(t)}.$$

Use the condition $u'(t)y_1(t) + v'(t)y_2(t) = 0$ introduced by Lagrange.

(b) Using part (a), find a general solution to

$$y'' - 3y' + 2y = \sin(e^{-t}).$$

Problem 107. (2015 Fall POSTECH Differential Equations). Use the properties of the Laplace transform.

(a) Find $f(t) = \mathcal{L}^{-1}\{F(s)\}(t)$ corresponding to

$$F(s) = \sum_{n=0}^{\infty} \frac{1}{s^{n+1}}, \qquad s > 1.$$

(b) Find $F(s) = \mathcal{L}\{f(t)\}(s)$ corresponding to

$$f(t) = e^t \frac{\mathrm{d}^n}{\mathrm{d}t^n}(t^n e^{-t}).$$

Problem 108. (2015 Fall POSTECH Differential Equations). Use the Laplace transform to solve the following problems.

(a) $y'' - 2y' - 3y = f(t)$, $y(0) = 1$, $y'(0) = 0$, where

$$f(t) = \begin{cases} 0, & 0 \le t < 4, \\ 1, & t \ge 4. \end{cases}$$

(b) $y'' + 9y = \cos 3t$, $y(0) = 0$, $y'(0) = 1$.

Problem 109. (2015 Fall POSTECH Differential Equations). Consider the equation

$$x^2y'' + 5xy' + (x+4)y = 0,$$

which has a regular singular point at $x = 0$.

(a) By Fuchs' theorem there is at least one solution of the form $y_1(x) = \sum_{n=0}^{\infty} a_n x^{r+n}$ ($x > 0$, $a_0 \neq 0$). Determine r.

(b) Show that the second solution is of the form

$$y_2(x) = y_1(x)\ln x + x^r \sum_{n=1}^{\infty} b_n x^n, \qquad x > 0,$$

where $y_1(x)$ and r are given in part (a).

Problem 110. (2016 Spring POSTECH Differential Equations). Express the solution of

$$y'' + 3y' + 2y = \delta(t-1), \qquad y(0) = 2, \quad y'(0) = -1,$$

in terms of a convolution integral.

Problem 111. (2016 Spring POSTECH Differential Equations). Find a fundamental matrix $\mathbf{\Phi}(t)$ for the system

$$\mathbf{x}' = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 0 & 5 \\ -1 & -1 & 4 \end{pmatrix} \mathbf{x}.$$

Problem 112. (2016 Spring POSTECH Differential Equations). Consider a uniform bar of length $L = 20$ and thermal diffusivity $\alpha^2 = 1$ with initial temperature $f(x) = 20 - x$ for $0 \leq x \leq 20$. Assume that the end at $x = 0$ is insulated, while the temperature at $x = 20$ is held at $0\,^\circ\mathrm{C}$.

(a) Find the fundamental solutions of this heat-conduction problem.

(b) Find the temperature $u(x,t)$ of the bar as a formal series.

Problem 113. (2016 Spring POSTECH Differential Equations).

(a) Using the change of variables $x = r\cos\theta$, $y = r\sin\theta$, compute and show that the Laplace operator $\Delta u = u_{xx} + u_{yy}$ becomes

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}.$$

(b) Find the solution $u(r,\theta)$ of Laplace's equation

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$$

outside the circle $r = a$ satisfying $u(a,\theta) = f(\theta)$, $0 \leq \theta \leq 2\pi$.

Problem 114. (2016 Spring POSTECH Differential Equations). Solve the nonhomogeneous boundary-value problem

$$\begin{cases} u_t = u_{xx} - 2, & 0 < x < 1,\ t > 0, \\ u(0,t) = 2, \quad u(1,t) = 0, \\ u(x,0) = x^2 - 3x + 3. \end{cases}$$

Problem 115. (2016 Spring POSTECH Differential Equations).

(a) Transform the equation into an equivalent self-adjoint form:

$$(1-x^2)y'' - xy' + \alpha^2 y = 0.$$

(b) Determine the normalized eigenfunctions of

$$y'' - 2y' + (1+\lambda)y = 0, \qquad y(0) = y(1) = 0.$$

Problem 116. (2016 Spring POSTECH Differential Equations).

(a) Suppose the functions $p(x), p'(x), q(x), r(x)$ are continuous and $p(x), r(x) > 0$. Let λ_n be an eigenvalue of the Sturm–Liouville problem

$$\begin{cases} (p(x)y')' - q(x)y + \lambda r(x)y = 0, & 0 < x < 1, \\ \alpha_1 y(0) + \alpha_2 y'(0) = 0, \\ \beta_1 y(1) + \beta_2 y'(1) = 0, \end{cases}$$

where the real numbers α_1, α_2 are not both zero, nor are β_1, β_2. If ϕ_1 and ϕ_2 are eigenfunctions corresponding to λ_n, show that $W(\phi_1, \phi_2)(0) = 0$.

(b) Show that the eigenfunction ϕ of the Sturm–Liouville problem is real-valued.

Problem 117. (2017 Spring POSTECH Differential Equations). Solve the following first-order differential equations.

(a) $y' + y^2 \cos t = 0$.

(b) $(6x^2 - 2xy + 4) + (6y^2 - x^2 + 2)y' = 0$.

(c) $ty' + 2y = 2\sin t$, $y(\pi/2) = 1$, for $t > 0$.

(d) $y' = \dfrac{y}{t} + \dfrac{t}{y}$, $t > 0$.

Problem 118. (2017 Spring POSTECH Differential Equations). (Mixing problem.) A tank that can contain 500 L of water originally contains 100 L of salt water in which 5 kg of salt is dissolved. Then water containing 2 kg of salt per L is poured into the tank at 5 L/min, and the well-stirred mixture is allowed to leave at 3 L/min. Determine the quantity of salt when the salt water begins to overflow.

Problem 119. (2017 Spring POSTECH Differential Equations). Solve the following second-order differential equations.

(a) $y'' + y' - 2y = 2t$, $y(0) = 0$, $y'(0) = 2$.

(b) $t^2y'' - 2y = 4t^2 - 3$, $t > 0$, given two homogeneous solutions $y_1(t) = t^2$ and $y_2(t) = 1/t$.

(c) $(x-1)^2y'' + 8(x-1)y' + 12y = 0$, $x > 1$.

Problem 120. (2017 Spring POSTECH Differential Equations). Find the general solution using the method of undetermined coefficients:

$$y'' + y = x + 2e^{-2x} + \sin x.$$

Problem 121. (2017 Spring POSTECH Differential Equations). Use the Laplace transform to solve the following problems.

(a) $y'' + 3y' + 2y = e^t$, $y(0) = 0$, $y'(0) = 2$.

(b) $y'' + y = g(t)$, $y(0) = 0$, $y'(0) = 2$, where

$$g(t) = \begin{cases} t/2, & 0 \le t < 6, \\ 3, & t \ge 6. \end{cases}$$

(c) $y^{(4)} - y = 0$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = 1$, $y^{(3)}(0) = 0$.

(d) Find $f(t)$ satisfying

$$f(t) = 2t^2 + \int_0^t f(t-\tau)e^{-2\tau}\,\mathrm{d}\tau.$$

Problem 122. (2017 Spring POSTECH Differential Equations). Consider the differential equation

$$xy'' + xy' + y = 0.$$

(a) Using the Laplace transform, find the solution satisfying $y(0) = 0$, $y'(0) = 1$. *Hint.* Use $\mathcal{L}\{xf(x)\}(s) = -\dfrac{\mathrm{d}}{\mathrm{d}s}\mathcal{L}\{f(x)\}(s)$.

(b) $y_1(x) = xe^{-x}$ is a solution of the differential equation. Using this and reduction of order, find a second series solution. Express the answer in the form $y_2 = f(x) + P(x)$, where $f(x)$ involves a logarithm and $P(x)$ is a power series about $x = 0$. Show the power-series expansion through the cubic term. *Hint.* Use the Taylor expansion.

Problem 123. (2017 Spring POSTECH Differential Equations). Find the first four nonzero terms of each of two linearly independent series solutions about $x = 0$ of

$$x^2y'' + x(x-2)y' + (x^2+2)y = 0, \qquad x > 0.$$

Problem 124. (2017 Spring POSTECH Differential Equations). Find the fundamental matrix $\mathbf{\Phi}(t)$ for the system

$$\mathbf{x}' = \begin{pmatrix} 1 & -4 \\ 4 & -7 \end{pmatrix}\mathbf{x}$$

satisfying $\mathbf{\Phi}(0) = \mathbf{I}$.

Problem 125. (2017 Spring POSTECH Differential Equations). Let $f(x) = x$ for $-1 < x < 1$, $f(1) = f(-1) = 0$, and $f(x+2) = f(x)$ for all x.

(a) Find the Fourier series of $f(x)$.

(b) Show that $\dfrac{\pi^2}{6} = \displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$.

Problem 126. (2017 Spring POSTECH Differential Equations). A metal rod with $\alpha^2 = 1$ and length 50 initially has a uniform temperature of 20 °C throughout. For $t > 0$, the end $x = 0$ is held at 10 °C, while the end $x = 50$ is held at 60 °C. Find the temperature $u(x, t)$ at position x and time t.

Problem 127. (2017 Spring POSTECH Differential Equations). Find the solution $u(x,t)$ of the wave equation

$$c^2 u_{xx} = u_{tt}, \qquad 0 < x < L, \quad t > 0,$$

subject to the boundary and initial conditions

$$u(0,t) = u(L,t) = 0, \quad t \geq 0, \qquad u_t(x,0) = 0, \quad u(x,0) = f(x), \quad 0 \leq x \leq L,$$

where f is a function on $\mathbb{R}$ with a continuous second derivative that is not identically zero. Express the solution $u(x,t)$ in terms of $f(x+ct)$ and $f(x-ct)$.

Problem 128. (2017 Spring POSTECH Differential Equations). Find a trigonometric series solution $u(x,y)$ of Laplace's equation

$$u_{xx} + u_{yy} = 0$$

in the rectangle $0 < x < 1$, $0 < y < 1$, with boundary conditions

$$u(x,0) = 0, \quad u(x,1) = e^x, \quad 0 \leq x \leq 1,$$

$$u(0,y) = 0, \quad u(1,y) = 0, \quad 0 \leq y \leq 1.$$

Explain why the series solution converges.

Problem 129. (2017 Spring POSTECH Differential Equations). Show that in polar coordinates (r,θ) Laplace's equation $u_{xx} + u_{yy} = 0$ becomes

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0.$$

Assume that u has continuous partial derivatives of all orders.

Problem 130. (2017 Spring POSTECH Differential Equations). Let $a > 0$. Find a bounded solution $u(r,\theta)$ of

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$$

in the polar region $0 < r < a$ with boundary condition $u(a,\theta) = f(\theta)$ for $0 \leq \theta < 2\pi$, where f is a continuous 2π-periodic function on $\mathbb{R}$ that is not identically zero.

Problem 131. (2017 Spring POSTECH Differential Equations). Consider the Sturm–Liouville problem

$$y'' - y + \lambda y = 0, \quad 0 < x < 1, \qquad y'(0) = 0, \quad y(1) = 0.$$

Define $L[y] = -y'' + y$. Show by direct computation that if $u(x)$ and $v(x)$ are twice differentiable functions on $(0,1)$ satisfying the boundary conditions, then

$$\int_0^1 L[u]v \, \mathrm{d}x = \int_0^1 L[v]u \, \mathrm{d}x.$$

Problem 132. (2017 Spring POSTECH Differential Equations). The Sturm–Liouville problem

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(1) + y'(1) = 0$$

has a sequence of real eigenvalues $(\lambda_n)_{n \geq 1}$ such that $0 < \lambda_1 < \lambda_2 < \cdots$.

(a) Show that λ_n must satisfy $\tan\sqrt{\lambda_n} = -\sqrt{\lambda_n}$.

(b) Show that the normalized eigenfunction $\phi_n(x)$ corresponding to λ_n is

$$\phi_n(x) = \frac{\sqrt{2}}{\left(1+\cos^2\sqrt{\lambda_n}\right)^{1/2}} \sin\left(\sqrt{\lambda_n}\, x\right).$$

(c) Find the coefficients a_n such that $2 = \sum_{n=1}^{\infty} a_n \phi_n(x)$ for all $x \in (0,1)$.

Problem 133. (2017 Spring POSTECH Differential Equations). Let $u(x,t)$ be a solution of

$$u_t - k u_{xx} = f(x,t)$$

subject to $u(0,t) = u(L,t) = 0$ for $t > 0$ and $u(x,0) = 0$ for $0 \le x \le L$. Show that if

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin\frac{n\pi x}{L}, \qquad f(x,t) = \sum_{n=1}^{\infty} c_n(t) \sin\frac{n\pi x}{L}$$

for $0 \le x \le L$, $t > 0$, then

$$b_n(t) = \int_0^t e^{-kn^2\pi^2(t-s)/L^2} c_n(s)\, ds.$$

Assume that the coefficients b_n and c_n decay sufficiently rapidly as $n \to \infty$ so that the series converge uniformly and can be differentiated term by term.

Problem 134. (2018 Spring POSTECH Differential Equations). Let $\mathbf{x}^{(1)}$, $\mathbf{x}^{(2)}$ and $\mathbf{x}^{(3)}$ in $\mathbb{R}^3$ be solutions of the homogeneous system

$$\mathbf{x}' = \mathbf{A}(t)\mathbf{x},$$

where each entry of the 3×3 matrix $\mathbf{A}(t)$ is continuous on an interval I. Let $\mathbf{X}(t)$ be the 3×3 matrix whose columns are $\mathbf{x}^{(k)}(t)$. Show that the Wronskian

$$W(t) = W(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \mathbf{x}^{(3)})(t) = \det \mathbf{X}(t)$$

satisfies

$$W'(t) = (a_{11}(t) + a_{22}(t) + a_{33}(t))W(t).$$

For the 3×3 case, you may use properties of determinants or direct calculation.

Problem 135. (2018 Spring POSTECH Differential Equations). Consider the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{pmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Solve the system and write a fundamental matrix $\mathbf{\Phi}(t)$.

Problem 136. (2018 Spring POSTECH Differential Equations). Find the general solution of

$$\mathbf{x}' = \begin{pmatrix} 3 & -18 \\ 2 & -9 \end{pmatrix} \mathbf{x}.$$

Problem 137. (2018 Spring POSTECH Differential Equations). Find two linearly independent real-valued solutions of

$$\mathbf{x}' = \begin{pmatrix} 1 & -1 \\ 5 & -3 \end{pmatrix} \mathbf{x}.$$

Problem 138. (2018 Spring POSTECH Differential Equations). Suppose that $f(x) = e^{2x}$ for $0 \le x \le 1$.

(a) Find the Fourier cosine series of the even periodic extension g_e of f with period 2.

(b) Find the Fourier sine series of the odd periodic extension g_o of f with period 2.

Problem 139. (2018 Spring POSTECH Differential Equations). Let $(\phi_n(x))_{n\ge 0}$ be an orthonormal system of real-valued functions on $[a, b]$ with the inner product

$$\langle f, g\rangle = \int_a^b f(x)g(x)\,\mathrm{d}x, \qquad \langle \phi_n, \phi_m\rangle = \delta_{nm}.$$

Assume continuity of the inner product: $\langle f, \lim_{n\to\infty} g_n\rangle = \lim_{n\to\infty}\langle f, g_n\rangle$.

(a) Prove that if $f = \sum_{n=0}^{\infty} a_n\phi_n$ and $g = \sum_{n=0}^{\infty} b_n\phi_n$, then

$$\langle f, g\rangle = \sum_{n=0}^{\infty} a_n b_n.$$

(b) Find the inner product $(f, g) = \dfrac{2}{L}\displaystyle\int_0^L f(x)g(x)\,\mathrm{d}x$ if

$$f = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}, \qquad g = \frac{b_0}{2} + \sum_{n=1}^{\infty} b_n \cos\frac{n\pi x}{L}.$$

Problem 140. (2018 Spring POSTECH Differential Equations). Solve the heat equation with a source:

$$u_{xx} + x = u_t, \qquad 0 < x < 1, \quad t > 0,$$

subject to $u(0, t) = u(1, t) = 0$ for $t > 0$ and $u(x, 0) = x$ for $0 < x < 1$.

Problem 141. (2018 Spring POSTECH Differential Equations). Suppose that $u(x, t)$ satisfies the wave equation

$$u_{xx} = u_{tt}, \qquad 0 < x < L, \quad t > 0.$$

Solve for $u(x, t)$ with the conditions

$$u(0, t) = u(L, t) = 0, \qquad u(x, 0) = 0, \quad u_t(x, 0) = x(L - x).$$

Problem 142. (2018 Spring POSTECH Differential Equations). Given $a > 0$ and $0 < L < \pi/2$, find the solution $u(r, \theta)$ of

$$\nabla^2 u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, \qquad 0 < r < a, \quad 0 < \theta < L,$$

subject to

$$u(0, 0) = 0, \quad u(r, 0) = 0, \quad u_\theta(r, L) = 0, \quad 0 < r < a,$$
$$u(a, \theta) = f(\theta), \qquad 0 < \theta < L.$$

Problem 143. (2018 Fall POSTECH Differential Equations). Consider the initial-value problem

$$y'' - 2y' + 2y = e^{-t}, \qquad y(0) = 1, \quad y'(0) = 1.$$

(a) Solve the equation without using the Laplace transform.
(b) Solve the equation using the Laplace transform.

Problem 144. (2018 Fall POSTECH Differential Equations). Let

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix}.$$

Find all solutions of $\mathbf{x}' = \mathbf{A}\mathbf{x}$.

Problem 145. (2018 Fall POSTECH Differential Equations). Let $u(x,t)$ be the temperature function of a uniform rod of length L with thermal diffusivity $\alpha^2 = k/(\rho s)$. Given an initial temperature $u(x,0) = \cos(\pi x/L)$ for $0 \le x \le L$, assume both ends of the rod are insulated: $u_x(0,t) = u_x(L,t) = 0$ for $t > 0$.

(a) Find the temperature $u(x,t)$.
(b) What is the steady-state temperature as $t \to \infty$?
(c) Describe briefly how the temperature in the rod changes as time progresses.

Problem 146. (2018 Fall POSTECH Differential Equations). Solve the initial-boundary value problem

$$u_{xx} - u_y = -Ae^{-\alpha x}, \qquad A \ge 0,\ \alpha > 0,$$

with conditions

$$u(0,y) = 0, \quad u(L,y) = 0, \quad y > 0, \qquad u(x,0) = f(x), \quad 0 < x < L.$$

Hint. Let $u(x,y) = v(x,y) + w(x,y)$ so that v satisfies the homogeneous equation

$$v_{xx} - v_y = 0$$

and homogeneous boundary conditions $v(0,y) = v(L,y) = 0$.

Problem 147. (2018 Fall POSTECH Differential Equations). Solve the initial-boundary value problem

$$\begin{cases} u_{tt} = c^2 u_{xx} + u, & x \in (0,1),\ t > 0, \\ u(0,t) = u(1,t) = 0, & t > 0, \\ u(x,0) = f(x), & x \in (0,1), \\ u_t(x,0) = 0, & x \in (0,1). \end{cases}$$

Problem 148. (2018 Fall POSTECH Differential Equations). Solve the system

$$\begin{cases} \dfrac{\mathrm{d}x}{\mathrm{d}t} - 6x + 3y = 8e^t, \\ \dfrac{\mathrm{d}y}{\mathrm{d}t} - 2x - y = 4e^t, \end{cases} \qquad x(0) = -1, \quad y(0) = 0.$$

Problem 149. (2018 Fall POSTECH Differential Equations). Consider the eigenvalue problem

$$\begin{cases} -y'' = \lambda y, & x \in (0,1), \\ y(0) - 3y'(0) = 0, \quad y'(1) = 0. \end{cases} \tag{1}$$

(a) Find the eigenvalues of (1).

(b) Determine the corresponding normalized eigenfunctions. *(Hint. A function ϕ is normalized if $\int_0^1 \phi^2(x)\,\mathrm{d}x = 1$.)*

Problem 150. (2018 Fall POSTECH Differential Equations). Solve the initial-value problem

$$\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t, \qquad \mathbf{x}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Problem 151. (2019 Spring POSTECH Differential Equations). Solve the following differential equations.

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y}{x} + 2\dfrac{x}{y}$, $y(1) = 2$.

(b) $(4xy^2 + 3y) + (2x^2y + x)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$.

Problem 152. (2019 Spring POSTECH Differential Equations). At time $t = 0$, a tank contains Q_0 grams of salt dissolved in 100 L of water. Assume that water containing γ grams of salt per liter is pumped into the tank at a rate of 5 L/min and the well-stirred mixture leaves the tank at the same rate. Find an expression for the amount $Q(t)$ of salt in the tank at time t in terms of γ and Q_0. Compute $\lim\limits_{t\to\infty} Q(t)$.

Problem 153. (2019 Spring POSTECH Differential Equations). Find all possible values of b for which the solution $y(t)$ of

$$y'' - 2by' + 4y = 0, \qquad y(0) = 0, \quad y'(0) = 1$$

remains finite as $t \to \infty$.

Problem 154. (2019 Spring POSTECH Differential Equations). Solve the following differential equations.

(a) $y'' - 2y' + y = 0$.

(b) $y'' - 2y' + y = e^t + 2\sin t$.

Problem 155. (2019 Spring POSTECH Differential Equations). Consider

$$t^2y'' - t(t+2)y' + (t+2)y = t^3, \qquad t > 0.$$

(a) Show that $y_1(t) = t$ is a solution of the corresponding homogeneous equation

$$t^2y'' - t(t+2)y' + (t+2)y = 0.$$

(b) Use reduction of order to find a second solution of the homogeneous equation.

(c) Use variation of parameters to solve

$$t^2y'' - t(t+2)y' + (t+2)y = 4t^3, \qquad y(1) = e, \quad y'(1) = 2e - 1.$$

Problem 156. (2019 Spring POSTECH Differential Equations). Find two series solutions of

$$y'' + (t-1)y' + 2y = 0$$

about $t_0 = 1$ which form a fundamental set of solutions. (You may use the notation $n!! = n(n-2)\cdots(n-2\ell)$, where $n - 2\ell = 1$ or 2; for example, $6!! = 6 \cdot 4 \cdot 2$ and $7!! = 7 \cdot 5 \cdot 3 \cdot 1$.)

Problem 157. (2019 Spring POSTECH Differential Equations). Solve the initial-value problem

$$2t^2y'' + ty' - 3y = 0, \qquad y(1) = 1, \quad y'(1) = 2.$$

Problem 158. (2019 Spring POSTECH Differential Equations). We seek Frobenius series solutions of the form $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ to

$$4xy'' + y' + xy = 0, \qquad x > 0.$$

(a) Determine the indicial equation and its roots.
(b) Write the first three nonzero terms of each solution.

Problem 159. (2019 Spring POSTECH Differential Equations).

(a) Find the inverse Laplace transform of $\dfrac{1}{(s^2+1)^2}$.
(b) Solve

$$y'' + y = \sin t + H(t-\pi)\sin(t-\pi), \qquad y(0) = 1, \quad y'(0) = 0.$$

Problem 160. (2019 Spring POSTECH Differential Equations). Let $y(t)$ solve

$$y'' + 2y' + 2y = \delta(t-\pi), \qquad y(0) = 1, \quad y'(0) = 3.$$

Find $y\left(\dfrac{3\pi}{2}\right)$.

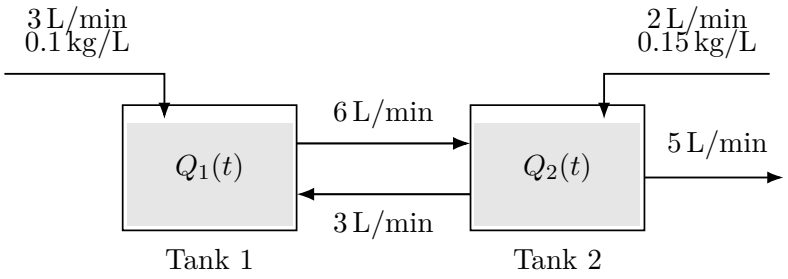
Problem 161. (2019 Spring POSTECH Differential Equations).

(a) Find the inverse Laplace transform of $\dfrac{1}{s^3-1}$.
(b) Find the solution of

$$y^{(3)} - y = g(t), \qquad y(0) = y'(0) = y''(0) = 0$$

in terms of a convolution integral.

Problem 162. (2019 Spring POSTECH Differential Equations). Consider the two interconnected tanks shown below.



Tank 1 initially contains 10 liters of water, and Tank 2 initially contains 5 liters of water. Water containing 0.1 kg/L of salt flows into Tank 1 at a rate of 3 L/min. The mixture in Tank 1 flows into Tank 2 at a rate of 6 L/min. At the same time, the mixture in Tank 2 flows into Tank 1 at a rate of 3 L/min and drains from Tank 2 at a rate of 5 L/min. Water containing 0.15 kg/L of salt also flows into Tank 2 at a rate of 2 L/min. Let $Q_1(t)$ and $Q_2(t)$ be the respective amounts of salt in each tank at time t.

(a) Write the differential equations modelling the flow process.

(b) Find Q_1 and Q_2 for which the system is in equilibrium (unchanging with time).

Problem 163. (2019 Spring POSTECH Differential Equations). Find the general solution of

$$\mathbf{x}' = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix} \mathbf{x}.$$

Problem 164. (2019 Spring POSTECH Differential Equations). Find all real numbers a such that the solution $\mathbf{x}(t)$ of

$$\mathbf{x}' = \begin{pmatrix} a & -1 \\ 1 & a \end{pmatrix} \mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

satisfies $\lim_{t\to\infty} \mathbf{x}(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Problem 165. (2019 Spring POSTECH Differential Equations).

(a) Solve the homogeneous system

$$\mathbf{x}' = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} \mathbf{x}.$$

(b) Solve the nonhomogeneous system

$$\mathbf{y}' = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} \mathbf{y} + \begin{pmatrix} 2 \\ -2 \end{pmatrix} e^t, \qquad \mathbf{y}(0) = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Problem 166. (2019 Spring POSTECH Differential Equations). Find the general solution of

$$\mathbf{x}' = \begin{pmatrix} -1 & 0 & 0 \\ -4 & -1 & 0 \\ 3 & 6 & 1 \end{pmatrix} \mathbf{x}.$$

Problem 167. (2019 Spring POSTECH Differential Equations). Solve the heat conduction problem

$$\begin{cases} 9u_{xx} = u_t, & 0 < x < 10,\ t > 0, \\ u_x(0,t) = 1, \quad u(10,t) = 0, & t > 0, \\ u(x,0) = 0, & 0 \le x \le 10. \end{cases}$$

(Hint. Seek a solution of the form $u(x,t) = v(x) + w(x,t)$.)

Problem 168. (2019 Spring POSTECH Differential Equations). Use separation of variables to solve

$$u_{tt} + u = u_{xx}, \qquad 0 < x < \pi, \quad t > 0,$$

subject to the boundary conditions $u(0,t) = u(\pi,t) = 0$ and the initial conditions $u(x,0) = 0$, $u_t(x,0) = f(x)$.

Problem 169. (2019 Spring POSTECH Differential Equations). Find the solution $u(x,y)$ of Laplace's equation in the rectangle $0 < x < 2$, $0 < y < 4$, with boundary conditions

$$\begin{aligned} u(0,y) = 0, \quad u(2,y) = f(y), \quad 0 < y < 4, \\ u(x,0) = 0, \quad u(x,4) = 0, \qquad 0 < x < 2, \end{aligned}$$

where $f(y) = (y-2)^2 - 4 = y^2 - 4y$.

Problem 170. (2019 Spring POSTECH Differential Equations). Find the solution $u(r,\theta)$ of Laplace's equation

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$$

in the circular sector $0 < r < 1$, $0 < \theta < \frac{\pi}{4}$, satisfying

$$\begin{aligned} u_\theta(r,0) = 0, \qquad u_\theta\left(r, \frac{\pi}{4}\right) = 0, \quad 0 < r < 1, \\ u(1,\theta) = f(\theta), \qquad\qquad 0 < \theta < \frac{\pi}{4}. \end{aligned}$$

Assume u is single-valued and bounded in the region.

Problem 171. (2019 Fall POSTECH Differential Equations). Solve the initial-value problem

$$y' = y + e^t, \qquad y(0) = 2.$$

Problem 172. (2019 Fall POSTECH Differential Equations). Consider the initial-value problem

$$(2 + x^2)y'' - xy' + 4y = 0, \qquad y(1) = 2, \quad y'(1) = 3.$$

(a) Find the longest interval on which the solution is certain to exist.

(b) Determine a lower bound for the radius of convergence of the power-series solution centered at $x = 1$.

Problem 173. (2019 Fall POSTECH Differential Equations). Consider the initial-value problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y^4 + 1, \qquad y(0) = 1.$$

What can you say about existence, uniqueness, and the interval on which the solution exists? You may use a theorem; be as precise and concise as possible.

Problem 174. (2019 Fall POSTECH Differential Equations). Consider the differential equation

$$xy'' + (1-x)y' + 2y = 0, \qquad x > 0.$$

(a) Explain why $x = 0$ is a singular point and is a regular one.
(b) Write the indicial equation and find its roots.
(c) Write the form of a first series solution $y_1(x)$ with undetermined coefficients (do not determine the coefficients, but determine the exponents). For what values of x does the series converge?
(d) Write the form of a second series solution $y_2(x)$ with undetermined coefficients (do not determine the coefficients). For what values of x does the series converge?
(e) Show that $y_1(x)$ and $y_2(x)$ are linearly independent.

Problem 175. (2019 Fall POSTECH Differential Equations). Consider the differential equation

$$y'' + 4y' + 4y = te^{-2t}.$$

(a) Let y_p be a particular solution, and y_1, y_2 be linearly independent solutions of the corresponding homogeneous equation. Write down the general solution in terms of y_1, y_2, y_p.
(b) Find y_1 and y_2.
(c) Write down a form for y_p using undetermined coefficients. Do not determine the coefficients.
(d) Use variation of parameters to find y_p.

Problem 176. (2019 Fall POSTECH Differential Equations). Find the general solution of

$$4x^2y'' + 8xy' + 17y = 0, \qquad x > 0.$$

Problem 177. (2019 Fall POSTECH Differential Equations). Solve the initial-value problem

$$(2x - y) + (2y - x)y' = 0, \qquad y(1) = 4.$$

Express the solution explicitly as a function of x.

Problem 178. (2019 Fall POSTECH Differential Equations). The function $y_1(x) = e^x$ is a solution of

$$(x-1)y'' - xy' + y = 0, \qquad x > 1.$$

Use reduction of order to find a second solution.

Problem 179. (2019 Fall POSTECH Differential Equations). Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be real-valued solutions of $\mathbf{x}' = \mathbf{A}\mathbf{x}$, where $\mathbf{A}$ is a real $n \times n$ matrix. Determine whether each statement is true or false. No justification is required.

(1) If $\mathbf{x}_1(1), \dots, \mathbf{x}_n(1) \in \mathbb{R}^n$ are linearly independent, then $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is a fundamental set of solutions.
(2) If $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is a fundamental set of solutions, then $e^{t\mathbf{A}} = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$.
(3) The columns of $e^{t\mathbf{A}}$ form a fundamental set of solutions.
(4) The general solution is $\mathbf{c}e^{t\mathbf{A}}$, where $\mathbf{c} \in \mathbb{R}^n$ is an arbitrary vector.
(5) $W(\mathbf{x}_1, \dots, \mathbf{x}_n)(t) = \det(e^{t\mathbf{A}})$.

Problem 180. (2019 Fall POSTECH Differential Equations). Transform

$$\begin{pmatrix} x''(t) \\ y''(t) \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

into a system of first-order equations. Do not solve it. Express the answer using matrices; define any newly introduced functions clearly.

Problem 181. (2019 Fall POSTECH Differential Equations). Solve the initial-value problem

$$4y'' + 4y' + 17y = g(t), \qquad y(0) = 0, \quad y'(0) = 1,$$

using the Laplace transform. Express the solution as a convolution integral.

Problem 182. (2019 Fall POSTECH Differential Equations). Find linearly independent real-valued solutions $\mathbf{u}(t)$ and $\mathbf{v}(t)$ of

$$\mathbf{x}' = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix} \mathbf{x}.$$

Problem 183. (2019 Fall POSTECH Differential Equations). Solve the initial-value problem

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}.$$

Express the solution as a column vector. *Hint.* First find the general solution and determine the coefficients.

Problem 184. (2019 Fall POSTECH Differential Equations). Let $f\colon \mathbb{R} \to \mathbb{R}$ be an odd function of class C^2 with period $2L$. Consider the boundary-value problem

$$\begin{cases} a^2 u_{xx} = u_{tt}, & 0 < x < L, \quad t > 0, \\ u(0,t) = u(L,t) = 0, & t \geq 0, \\ u(x,0) = f(x), \quad u_t(x,0) = 0, & 0 \leq x \leq L. \end{cases}$$

(a) Write down a series solution and determine its coefficients. No justification is required.
(b) Show that the series in part (a) satisfies $a^2 u_{xx} = u_{tt}$.

Problem 185. (2019 Fall POSTECH Differential Equations). A 20-cm-long bar is made of a material with thermal diffusivity $\alpha^2 = 3$. Initially it is at a uniform temperature of 100 °C. At $t = 0$, the end $x = 0$ is cooled to 50 °C, while the end $x = 20$ is heated to 150 °C; thereafter, both ends remain at those temperatures.

(a) Set up a boundary-value problem (differential equation and boundary conditions).
(b) Using separation of variables and Fourier series, derive a series solution. Write formulas for the coefficients without computing their numerical values. You need not justify convergence or term-by-term differentiation.

Problem 186. (2019 Fall POSTECH Differential Equations). The electrostatic potential $\varphi(r,\theta)$ inside a circular disk $r \le a$ is a solution of

$$\begin{cases} \nabla^2\varphi = 0, & 0 < r < a, \quad 0 < \theta < 2\pi, \\ \varphi(a,\theta) = f(\theta), & 0 < \theta < 2\pi. \end{cases}$$

Using separation of variables and Fourier series, derive the series solution

$$\varphi(r,\theta) = \frac{c_0}{2} + \sum_{n=1}^{\infty} r^n (c_n \cos n\theta + k_n \sin n\theta).$$

Write formulas for the coefficients without computing numerical values. You need not justify convergence or term-by-term differentiation.

Problem 187. (2020 Spring POSTECH Differential Equations). Find a nonconstant solution of

$$(x^2y + y^3) + (x^3 + 2x^2y - 3xy^2)y' = 0.$$

Problem 188. (2020 Spring POSTECH Differential Equations). Let $b \ge 0$, and suppose $y = \phi(t)$ solves

$$y'' - by' + 4y = \sin t, \qquad y(0) = y'(0) = 0.$$

If $\phi(t)$ remains finite as $t \to \infty$, find b and $\phi(t)$.

Problem 189. (2020 Spring POSTECH Differential Equations). We seek solutions by the Frobenius method in the form $y = \sum_{n=0}^{\infty} a_n x^{n+r}$ to

$$5xy'' + y' + xy = 0, \qquad x > 0.$$

(a) Determine the indicial equation and its roots.
(b) Write the first three nonzero terms of each of two solutions.

Problem 190. (2020 Spring POSTECH Differential Equations). There are 78 stairs along the POSTECH school road from Jigok restaurant. Let $f(x)$ denote the function describing these 78 stairs.

(a) Express $f(x)$ in terms of the unit-step or Heaviside function.
(b) Find the Laplace transform $F(s)$ of $f(x)$.
(c) Solve the initial-value problem

$$u''(x) + 4u(x) = f(x), \qquad x > 0, \quad u(0) = 0, \quad u'(0) = 0.$$

Problem 191. (2020 Spring POSTECH Differential Equations). Find the general solution of

$$\mathbf{x}' = \begin{pmatrix} 3 & 5 & 0 \\ 0 & 0 & 1 \\ -2 & 0 & -1 \end{pmatrix} \mathbf{x}.$$

Problem 192. (2020 Spring POSTECH Differential Equations). Consider Laplace's equation $u_{xx} + u_{yy} = 0$ on the rectangle $R = (0,1) \times (0,1)$, with boundary conditions

$$u(x,0) = 0, \quad u(x,1) = e^{-x}, \quad 0 < x < 1,$$
$$u(0,y) = 0, \quad u(1,y) = 0, \qquad 0 < y < 1.$$

(a) Find $u(x,y)$ satisfying the boundary-value problem.
(b) Show that the solution satisfies the equation and boundary conditions.
(c) Discuss convergence of the solution series.

Problem 193. (2020 Fall POSTECH Differential Equations). Solve $y' = ty + t$, $y(0) = 0$.

Problem 194. (2020 Fall POSTECH Differential Equations). Consider $y'' - 2y' + 2y = 0$. Find a complex-valued solution and then use it to find two linearly independent real-valued solutions.

Problem 195. (2020 Fall POSTECH Differential Equations). Find the general solution of $x^2y'' - 2xy' + 2y = 0$.

Problem 196. (2020 Fall POSTECH Differential Equations). Consider

$$(y \sin x + x) + (y - \cos x)\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

(a) Determine whether this differential equation is exact.
(b) Find a solution satisfying $y(0) = 1$. (You may give it in implicit form.)

Problem 197. (2020 Fall POSTECH Differential Equations). Consider

$$(x^2 + 2)y'' + (x^3 + 3)y' + (x^4 + 4)y = 0, \qquad y(1) = 2, \quad y'(1) = 3.$$

(a) Find the longest interval on which you are certain that a solution exists.
(b) Find the longest interval on which you are certain that the power-series solution centered at $x = 1$ converges.

Problem 198. (2020 Fall POSTECH Differential Equations).

(a) Find the integral equation equivalent to $y' = t + y^2$, $y(0) = 0$.
(b) Find the first three Picard iterates $\phi_0(t)$, $\phi_1(t)$ and $\phi_2(t)$ approximating the solution.

Problem 199. (2020 Fall POSTECH Differential Equations). Solve $y'' - 6y' + 9y = e^{3t}$, $y(0) = 0$, $y'(0) = 1$.

Problem 200. (2020 Fall POSTECH Differential Equations). Find the general solution of $y'' + 4y = \cot t$ for $0 < t < \pi$.

Problem 201. (2020 Fall POSTECH Differential Equations). Consider

$$t^2y'' - t(t+3)y' + (t+3)y = 0, \qquad t > 0.$$

(a) Find a first solution y_1 by guessing.
(b) Use y_1 to find a second solution y_2. Simplify your answer so that it contains no definite integrals.

Problem 202. (2020 Fall POSTECH Differential Equations).

(a) Verify that $x = 0$ is a regular singular point of

$$2x^2y'' - 3xy' + (2 + x)y = 0.$$

(b) Find two series solutions y_1 and y_2. Check that they form a fundamental set of solutions and write the general solution.

Problem 203. (2020 Fall POSTECH Differential Equations). Find the solution of

$$y'' + 2y' + 3y = \delta(t - 2), \qquad y(0) = 0, \quad y'(0) = 0.$$

Problem 204. (2020 Fall POSTECH Differential Equations). Find the inverse Laplace transform of

$$F(s) = \frac{1}{(s + 1)(s^2 + 4)}$$

in two ways:

(a) Using partial fractions.

(b) Evaluating a convolution integral. *(Suggestion. Integrate by parts twice.)*

Problem 205. (2020 Fall POSTECH Differential Equations). Let $g\colon \mathbb{R} \to \mathbb{R}$ be piecewise continuous with $|g(t)| \le e^t$ for sufficiently large t. Solve using the Laplace transform

$$y''' + 3y'' + 3y' + y = g(t), \qquad y(1) = y'(1) = y''(1) = 0.$$

Problem 206. (2020 Fall POSTECH Differential Equations). Consider the system

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2\cos t \\ 2\sin t \end{pmatrix}.$$

(a) Suppose $\mathbf{x} = \mathbf{u} + i\mathbf{v}$ is a complex-valued solution with $\mathbf{u}$ and $\mathbf{v}$ real-valued. Is $\mathbf{u}$ a solution? Is $\mathbf{v}$ a solution? Justify your answers.

(b) Use diagonalization to find a real-valued particular solution. Your final answer must not contain i.

Problem 207. (2020 Fall POSTECH Differential Equations). Consider

$$\mathbf{x}' = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \mathbf{x}.$$

(a) Find a fundamental matrix explicitly.

(b) Find the solution satisfying $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, computing it explicitly.

Problem 208. (2020 Fall POSTECH Differential Equations).

(a) Find the Fourier series of $f(x) = x$ for $-\pi < x < \pi$, extended 2π-periodically to $\mathbb{R}$. What value of $f(\pi)$ makes the Fourier series converge to $f(x)$ for every $x \in \mathbb{R}$?

(b) Use part (a) to find $S = 1 - \dfrac{1}{3} + \dfrac{1}{5} - \dfrac{1}{7} + \cdots$.

Problem 209. (2020 Fall POSTECH Differential Equations). Find $u(x, y)$ satisfying Laplace's equation in $0 < x < 1$, $0 < y < 1$ and

$$\begin{aligned} u(0,y) = 0, \quad u(1,y) = 0, \qquad & 0 < y < 1, \\ u(x,0) = 0, \quad u(x,1) = f(x), \quad & 0 < x < 1, \end{aligned}$$

where $f(x) = \sin(3\pi x)$. Give a detailed argument and simplify the final answer as much as possible.

Problem 210. (2020 Fall POSTECH Differential Equations). A metal bar 10 cm long has thermal diffusivity $\alpha^2 = 1$ and is initially at a uniform temperature of 0°C. At $t = 0$ the end $x = 0$ is insulated, while the end $x = 10$ is heated to 100°C.

(a) Without calculation, find the steady-state temperature distribution at thermodynamic equilibrium.

(b) Estimate the temperature at $x = 5$ when $t = 40$.

Problem 211. (2020 Fall POSTECH Differential Equations). A string of length 3 satisfies

$$u_{xx} = u_{tt}, \qquad 0 < x < 3, \quad t > 0.$$

Its ends are fixed, and it is released from initial displacement $f(x)$ with zero initial velocity. Assume that f' is piecewise continuous (and f is continuous).

(a) Write down a formal series solution $u(x, t)$, determining its coefficients. You need not derive the series.

(b) Justify that the series from part (a) solves the boundary-value problem. In particular, show that it converges and satisfies the wave equation and boundary conditions. (Term-by-term differentiation of a trigonometric series is not always permitted.)

Problem 212. (2021 Fall POSTECH Differential Equations). Find a solution of

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x - y}.$$

(Suggestion. Substitute $v = y/x$ to obtain a familiar type of differential equation.)

Problem 213. (2021 Fall POSTECH Differential Equations). Find an integrating factor and solve

$$(y^2 + 4ye^x) + (2y + 2e^x)\frac{\mathrm{d}y}{\mathrm{d}x} = 0, \qquad y(0) = 1.$$

Problem 214. (2021 Fall POSTECH Differential Equations). A tank initially contains 24 liters of fresh water. Beginning at $t = 0$, water containing 2 g of salt per liter enters at a rate of 4 liters per minute, while the well-stirred mixture leaves at the same rate. Find the amount of salt in the tank at time t.

Problem 215. (2021 Fall POSTECH Differential Equations).

(a) State the integral equation equivalent to

$$y' = y^2 + 2t, \qquad y(0) = 0.$$

(b) Find the first three Picard iterates $\phi_0(t)$, $\phi_1(t)$, $\phi_2(t)$ approximating $y = \phi(t)$.

Problem 216. (2021 Fall POSTECH Differential Equations). Solve $y'' + 9y = \cos 3t$, $y(0) = y'(0) = 0$.

Problem 217. (2021 Fall POSTECH Differential Equations). Find the general solution of $y'' - 4y' + 4y = e^{2t}$.

Problem 218. (2021 Fall POSTECH Differential Equations). Find a particular solution of $y'' + y = \tan t$ on $0 < t < \frac{\pi}{2}$. (Recall $\dfrac{\mathrm{d}}{\mathrm{d}t} \ln|\sec t + \tan t| = \sec t$.)

Problem 219. (2021 Fall POSTECH Differential Equations). Consider

$$t^2 y'' - t(t+2)y' + (t+2)y = 0, \qquad t > 0.$$

(a) Find a first solution y_1 by guessing. Try simple functions.

(b) Use y_1 to find a second solution y_2. Simplify your answer so that it contains no definite integrals.

Problem 220. (2021 Fall POSTECH Differential Equations).

(a) Verify that $x = 0$ is a regular singular point of

$$2x^2 y'' - 3xy' + (2 + x)y = 0.$$

(b) Find two series solutions y_1 and y_2. Check that they form a fundamental set and state the general solution.

Problem 221. (2021 Fall POSTECH Differential Equations). Find the solution of

$$y'' + y' - 2y = \delta(t-2), \qquad y(0) = 1, \quad y'(0) = 0.$$

Problem 222. (2021 Fall POSTECH Differential Equations). Find the solution, in terms of a convolution integral, of

$$y'' + 4y = g(t), \qquad y(0) = 0, \quad y'(0) = 1.$$

Problem 223. (2021 Fall POSTECH Differential Equations). Find the inverse Laplace transform of

$$F(s) = \frac{1}{(s+3)(s^2+4)}$$

in two ways:

(a) By using partial fractions.

(b) By evaluating a convolution integral. *(Suggestion. Integrate by parts twice.)*

Problem 224. (2021 Fall POSTECH Differential Equations). Let $\mathbf{A} = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$.

(a) Find an invertible matrix $\mathbf{T}$ such that $\mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \mathbf{D}$ with $\mathbf{D}$ diagonal.

(b) Compute $e^{t\mathbf{A}}$.

(c) Find a fundamental matrix $\mathbf{\Phi}(t)$ satisfying $\mathbf{\Phi}(0) = \mathbf{I}$.

(d) Find the solution of $\mathbf{X}' = \mathbf{A}\mathbf{X}$ with $\mathbf{X}(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

Problem 225. (2021 Fall POSTECH Differential Equations). Consider

$$\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{g}(t) = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} e^t \\ 2 \end{pmatrix}.$$

Find the Jordan form $\mathbf{J}$ of $\mathbf{A}$ and use it to find the general solution. *(Suggestion. Find an invertible* $\mathbf{T}$ *such that* $\mathbf{T}^{-1}\mathbf{A}\mathbf{T} = \mathbf{J}$*.)*

Problem 226. (2021 Fall POSTECH Differential Equations).

(a) Find the Fourier series of $f(x) = x$ for $-1 < x < 1$, extended 2-periodically to $\mathbb{R}$. Define $f(1)$ so that its Fourier series converges to $f(x)$ everywhere.

(b) Use part (a) to evaluate $S = 1 - \dfrac{1}{3} + \dfrac{1}{5} - \dfrac{1}{7} + \cdots$.

Problem 227. (2021 Fall POSTECH Differential Equations). A metal rod of length 20 has thermal diffusivity $\alpha^2 = 1$. Its initial temperature distribution is $f(x) = x$. The boundary temperatures are $u(0,t) = 10$ and $u(20,t) = 0$ for $t > 0$. Find $u(x,t)$ in the rod.

Problem 228. (2021 Fall POSTECH Differential Equations). Find $u(x,y)$ satisfying Laplace's equation in the square $0 < x < 2$, $0 < y < 2$ and

$$\begin{aligned} u(0,y) &= 0, \quad u(2,y) = 0, \qquad 0 < y < 2, \\ u(x,0) &= 0, \quad u(x,2) = f(x), \quad 0 < x < 2, \end{aligned}$$

where $f(x) = \sin(4\pi x)$. Give a detailed argument and simplify the final answer as much as possible.

Problem 229. (2022 Spring POSTECH Differential Equations). Determine whether each statement is true or false. No justification is required.

(a) There exists a second-order linear homogeneous differential equation, whose coefficient functions are continuous on all of $\mathbb{R}$, with a fundamental set of solutions consisting of $y_1(x) = x^2$ and $y_2(x) = x^4$.

(b) Consider $2y'' + 4y' + 6.5y = 0$. Let $x_1 = y$ and $x_2 = y'$ to obtain a first-order system. The origin of the resulting phase portrait is a spiral point.

(c) A population y is modelled by $\dfrac{\mathrm{d}y}{\mathrm{d}t} = -\dfrac{1}{2500}(y^3 - 55y^2 + 250y)$. Then $y(t) = 5$ is a solution and is asymptotically stable.

(d) Suppose $y' = f(t,y)$, $y(0) = 0$, with f continuous on $R = \{(t,y) \mid |t| \le 1,\ |y| \le 1\}$ and $|f(t,y)| \le 0.5$ on R. Then a solution exists and is defined on $(-1,1)$.

(e) A solution to $y' = \sqrt{|y|}$, $y(0) = 0$, is unique.

Problem 230. (2022 Spring POSTECH Differential Equations). Solve the initial-value problem and find the interval of existence of the solution:

$$y' + \frac{4}{x}y = x^3y^2, \qquad y(2) = -1, \quad x > 0.$$

(Hint. You may use $2e^{-1/16} \approx 1.8788$*, or proceed without it.)*

Problem 231. (2022 Spring POSTECH Differential Equations). Consider $M(x,y) + N(x,y)y' = 0$, with

$$M(x,y) = \frac{-y}{x^2+y^2}, \qquad N(x,y) = \frac{x}{x^2+y^2}.$$

(a) Show that $M_y(x,y) = N_x(x,y)$ on $\mathbb{R}^2 \setminus \{(0,0)\}$.

(b) Nevertheless, the solution of the differential equation does not exist. Explain why.

(c) Restrict the domains of M and N to $D = \{(x,y) \in \mathbb{R}^2 \mid x > 0,\ y > 0\}$. Find the solution of the differential equation on D.

Problem 232. (2022 Spring POSTECH Differential Equations). Find the general solution of

$$(1+x^2)y'' + xy' - y = 0.$$

(Hint. What is meant by an exact second-order linear differential equation?)

Problem 233. (2022 Spring POSTECH Differential Equations). Find a particular solution of the nonhomogeneous system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t) = \begin{pmatrix} -3 & 1 \\ -1 & -5 \end{pmatrix}\mathbf{x} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} te^{-4t}.$$

Problem 234. (2022 Spring POSTECH Differential Equations). Two tanks initially contain 100 gallons of water each, with 80 lb of salt dissolved in Tank T_1 and 20 lb in Tank T_2. The liquid is stirred to remain uniform. Fresh water enters T_1 at 9.6 gal/min; liquid flows from T_1 to T_2 at 10 gal/min and from T_2 to T_1 at 0.4 gal/min; liquid drains out of T_2 at 9.6 gal/min. Find the masses of salt $y_1(t)$ and $y_2(t)$ in T_1 and T_2.

Problem 235. (2022 Spring POSTECH Differential Equations). Determine whether each statement is true or false. No justification is required.

(a) Let $D \subset \mathbb{R}^2$ be a closed set with simply connected boundary and open interior $D \setminus \partial D$. If $u\colon D \to \mathbb{R}$ is harmonic, then both its maximum and minimum occur only on the boundary of D.

(b) The radius of convergence of the Taylor series for $\dfrac{1}{1-x}$ about $x = 5$ is 5.

(c) The Laplace transform of $\dfrac{\sin x}{x}$ equals $\dfrac{\pi}{2} - \arctan s$. *(Hint.* $\mathcal{L}\{-xf(x)\}(s) = F'(s)$ *if* $F(s) = \mathcal{L}\{f(x)\}(s)$*.)*

(d) Suppose f and f' are continuous for $t \geq 0$ and are of exponential order at infinity. If $F(s) = \mathcal{L}\{f(x)\}(s)$, then $\lim\limits_{s\to\infty} F(s) = 0$.

Problem 236. (2022 Spring POSTECH Differential Equations). Show that

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots .$$

(Hint. Consider the Fourier series of $f(x) = x^2$ for $-1 \le x \le 1$, with $f(x+2) = f(x)$.)

Problem 237. (2022 Spring POSTECH Differential Equations). Consider a bar of length L. Its end at $x = L$ is maintained at 0°C, and its end at $x = 0$ is insulated. If $u(x,t)$ is the temperature at position x and time t, with initial temperature $u(x,0) = \dfrac{L-x}{L}$ and thermal diffusivity $\alpha^2 = 1$, find $u(x,t)$.

Problem 238. (2022 Spring POSTECH Differential Equations). Let $D = \{(x,y) \in \mathbb{R}^2 \mid x \ge 0,\ 0 \le y \le 1\}$. Find a solution $u(x,y)$ of Laplace's equation on D satisfying

$$\begin{cases} u(0,y) = y(1-y), \\ u_y(x,0) = 0, \\ u_y(x,1) = 0, \end{cases} \qquad \text{and} \qquad \lim_{x\to\infty} u(x,y) = 0.$$

Problem 239. (2022 Spring POSTECH Differential Equations). Consider

$$xy'' + (2-2x)y' + (x-2)y = 0.$$

Show that $x = 0$ is a regular singular point and find the general solution.

Problem 240. (2022 Spring POSTECH Differential Equations). Consider

$$\begin{cases} 2x' + y' - y = t, \\ x' + y' = t^2, \end{cases} \qquad x(0) = 1, \quad y(0) = 0.$$

(a) Apply the Laplace transform to the system, writing $X(s) = \mathcal{L}\{x(t)\}(s)$ and $Y(s) = \mathcal{L}\{y(t)\}(s)$.
(b) Solve the transformed equations for $X(s)$ and $Y(s)$.
(c) Find the inverse transforms of $X(s)$ and $Y(s)$.

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Problem 241. (2021 Spring KAIST Differential Equations and Applications). Answer the following questions.

(a) Solve the initial-value problem

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(50-y)(y-10)}{40}, \qquad y(0) = 18.$$

(b) Sketch the solution curve.
(c) Find the point on the solution curve where $\mathrm{d}y/\mathrm{d}x$ is maximized.

Problem 242. (2021 Spring KAIST Differential Equations and Applications). Solve the differential equation

$$(x+1)y' + (x+2)y = 2xe^{-x}, \qquad x > -1.$$

Problem 243. (2021 Spring KAIST Differential Equations and Applications). Solve the initial-value problem

$$y^2\cos(xy)\,\mathrm{d}x - (2 - 2\sin(xy) - xy\cos(xy))\,\mathrm{d}y = 0, \qquad y(\pi) = 2.$$

Problem 244. (2021 Spring KAIST Differential Equations and Applications). Find the general solution of the differential equation

$$y'' - \frac{1}{x}y' = -x\sin x, \qquad x > 0.$$

Problem 245. (2021 Spring KAIST Differential Equations and Applications). Consider the population model for a species of fish in a lake

$$\frac{\mathrm{d}P}{\mathrm{d}t} = 2P - \frac{P^2}{50},$$

where P is measured in thousands of fish and t is measured in years. The managing office of the lake wants to issue fishing licenses for the harvesting of some of the fish; this amounts to a constant term being subtracted off of the right-hand side of the above equation, which is a function of h, the number of licenses issued. Each fishing license is valid for the annual take of 3,000 fish. Answer the following questions.

(a) What is the largest number of licenses that can be issued if the goal is to keep a stable population of fish in the lake over the long term? The number of licenses is, of course, a nonnegative integer.

(b) If the largest number of licenses is actually issued, what is the expected long-term stable population of fish in the lake?

Problem 246. (2021 Spring KAIST Differential Equations and Applications). Consider the linear differential equation

$$ay'' + by' + cy = g(t),$$

where a, b, c are positive constants and $g(t)$ is a continuous function. If $y_1(t)$ and $y_2(t)$ are both solutions of the above equation, show that

$$\lim_{t\to\infty}(y_1(t) - y_2(t)) = 0.$$

Is this still true if $b = 0$?

Problem 247. (2021 Spring KAIST Differential Equations and Applications). Verify that $y_1(t) = 1$ and $y_2(t) = \sqrt{t}$ both solve the differential equation

$$yy'' + (y')^2 = 0, \qquad t > 0,$$

but $y(t) = c_1 + c_2\sqrt{t}$ is not a general solution of the differential equation. Explain why this result does not contradict the superposition principle for homogeneous equations we have learned in the class.

Problem 248. (2021 Spring KAIST Differential Equations and Applications). Find an integer n such that $\phi(x) = \sin(x^n)$ satisfies the differential equation

$$xy'' - y' + 4x^3 y = 0, \qquad x > 0.$$

Find the general solution of the equation.

Problem 249. (2021 Spring KAIST Differential Equations and Applications). Find a particular solution of the differential equation

$$y'' - 2y' + y = e^x \arctan x.$$

Problem 250. (2021 Spring KAIST Differential Equations and Applications). Using the method of Green's functions, find the solution of the initial-value problem

$$y'' - 10y' + 25y = e^{5t}, \qquad y(0) = -1, \qquad y'(0) = 1.$$

Problem 251. (2021 Spring KAIST Differential Equations and Applications). Find a function ϕ such that the function $y = \phi(\ln x)$ satisfies the differential equation

$$x^3 y''' + 2x^2 y'' - xy' + y = \frac{1}{x}, \qquad x > 0.$$

Find the general solution of the equation.

Problem 252. (2021 Spring KAIST Differential Equations and Applications). Find the general solution of the system of second-order differential equations

$$\begin{cases} x'' = -2x + y + t - 1, \\ y'' = 4x - 2y + 3 + \cos(2t). \end{cases}$$

Problem 253. (2021 Spring KAIST Differential Equations and Applications). For a piecewise continuous function $f(t)$ defined on $[0, \infty)$, we have studied in class that a sufficient condition guaranteeing the existence of the Laplace transform $\mathcal{L}\{f(t)\}(s)$ is that f be of exponential order. Meanwhile, there may be a piecewise continuous function that possesses the Laplace transform but is not of exponential order. It is known that the function

$$g(t) = 2te^{t^2} \cos(e^{t^2})$$

is not of exponential order. Nevertheless, show that $\mathcal{L}\{g(t)\}(s)$ exists. *(Hint. Use integration by parts.)*

Problem 254. (2021 Spring KAIST Differential Equations and Applications). For each $t \geq 0$, let

$$f(t) = \int_0^\infty \frac{\cos(t\eta)}{1 + \eta^2} \, \mathrm{d}\eta.$$

Show that

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty \frac{s}{(1 + \eta^2)(s^2 + \eta^2)} \, \mathrm{d}\eta = \frac{\pi}{2(s + 1)}.$$

(Hint. Consider changing the order of integration.)

Problem 255. (2021 Spring KAIST Differential Equations and Applications). Find the series solution of the initial-value problem

$$y'' + x^2y' + y = 0, \qquad y(0) = 2, \qquad y'(0) = 3,$$

up to the first four nonzero terms.

Problem 256. (2021 Spring KAIST Differential Equations and Applications). Find the general solution of the differential equation

$$4x^2y'' - 8x^2y' + (4x^2 + 1)y = 0.$$

Note that $x = 0$ is a singular point of the differential equation. *(Hint. When you obtain a three-term recurrence relation, consider an appropriate substitution that converts the three-term relation into a two-term relation.)*

Problem 257. (2021 Spring KAIST Differential Equations and Applications). Solve

$$\mathbf{x}' = \begin{pmatrix} -3 & -1 & -5 \\ -2 & 1 & -2 \\ 2 & 1 & 4 \end{pmatrix} \mathbf{x}$$

by diagonalization.

Problem 258. (2021 Spring KAIST Differential Equations and Applications). Find the general solution of the nonhomogeneous linear system

$$\mathbf{x}' = \begin{pmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2t+1 \\ 3 \\ 0 \end{pmatrix}.$$

Problem 259. (2021 Spring KAIST Differential Equations and Applications). Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 4 & 0 \end{pmatrix}.$$

(a) Find $e^{t\mathbf{A}}$.

(b) Solve the initial-value problem

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Problem 260. (2021 Spring KAIST Differential Equations and Applications). Consider the linear system

$$\mathbf{x}' = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} \mathbf{x}.$$

(a) Decide whether the origin is a stable critical point, an asymptotically stable critical point, or an unstable critical point for the system.

(b) Find the general solution.

(c) Find the solution $\mathbf{x}(t)$ such that

$$\mathbf{x}(0) = \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}.$$

Problem 261. (2021 Spring KAIST Differential Equations and Applications). Find the solution of the system

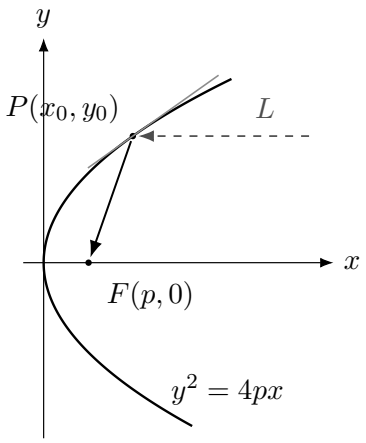
$$\begin{cases} x' = -5x + 4y, \\ y' = -2x + 4y, \end{cases}$$

and draw the phase plane near the critical point $(0, 0)$. Include the direction of some solution curves.

Problem 262. (2021 Spring KAIST Differential Equations and Applications). Find all the critical points of the following nonlinear system of differential equations and classify (if possible) whether they are stable (unstable) nodes or saddle points, spirals, etc.:

$$\begin{cases} x' = 0.2x(1 - 0.2x - 0.3y), \\ y' = 0.5y(1 - 0.1x - 0.2y). \end{cases}$$

Problem 263. (2021 Spring KAIST Differential Equations and Applications). Parabolas have the reflective property that whenever an incident ray parallel to the axis of a parabola is reflected on the parabola, then it will pass through the focus; the following figure depicts a case of the parabola $y^2 = 4px$ $(p > 0)$.



If we model such a reflective property as a differential equation, then we obtain the first-order nonlinear differential equation

$$y(y')^2 - 2(p - x)y' - y = 0.$$

More precisely, a solution $y = f(x)$ of the above differential equation is intended to satisfy the following: if an incident ray L parallel to the x-axis is reflected at the point $P(x_0, y_0)$ on the graph of $y = f(x)$, then it passes through the point $F(p, 0)$.

(a) Verify that the parabola $y^2 = 4px$ satisfies the given differential equation.

(b) Find a one-parameter family of solutions of the given differential equation. *(Hint. Use the substitution $w = y^2$. Then differentiate the modified equation with respect to the independent variable x.)*

(c) Characterize a common geometric property of the one-parameter family of solutions you obtained.

Problem 264. (2022 Spring KAIST Differential Equations and Applications). Solve the differential equation

$$y' + y = 1.$$

Problem 265. (2022 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$y' = y^2, \qquad y(0) = 1.$$

(a) Solve the initial-value problem.

(b) Find the maximal possible interval on which the solution $y(x)$ satisfying $y(0) = 1$ exists.

Problem 266. (2022 Spring KAIST Differential Equations and Applications). Consider a solution $y(x)$ of the autonomous differential equation

$$y' = y(y - 1).$$

(a) Find all constant solutions.

(b) Sketch the graph of the solution $y(x)$, $-\infty < x < \infty$, such that $y(0) = \frac{1}{2}$. You do not need to find the formula for $y(x)$.

(c) Let $y(x)$ be the solution such that $y(0) = \frac{1}{2}$. Find

$$\lim_{x \to -\infty} y(x) \qquad \text{and} \qquad \lim_{x \to +\infty} y(x).$$

You do not need to find the formula for $y(x)$.

(d) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = 2$. You do not need to find the explicit formula for $y(x)$.

(e) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = -1$. You do not need to find the explicit formula for $y(x)$.

Problem 267. (2022 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$(x^2 + y^2)y' + (3x^2 + 2x)y = -y^3.$$

(a) Rewrite the given differential equation as

$$M(x, y)\, \mathrm{d}x + N(x, y)\, \mathrm{d}y = 0$$

for some functions $M(x, y)$ and $N(x, y)$.

(b) Show that

$$\frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} = 3N(x, y).$$

(c) Let $\mu(x) = e^{3x}$. Show that

$$\frac{\partial(\mu(x)M(x, y))}{\partial y} - \frac{\partial(\mu(x)N(x, y))}{\partial x} = 0.$$

(d) Consider the equation

$$\mu(x)M(x, y)\, \mathrm{d}x + \mu(x)N(x, y)\, \mathrm{d}y = 0,$$

and find $f(x, y)$ satisfying

$$\frac{\partial f(x, y)}{\partial x} = \mu(x)M(x, y), \qquad \frac{\partial f(x, y)}{\partial y} = \mu(x)N(x, y).$$

(e) Express the solution of the given differential equation as

$$f(x, y) = C,$$

where C is an arbitrary constant.

Problem 268. (2022 Spring KAIST Differential Equations and Applications). Solve the initial-value problem

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos^2(x - y), \qquad y(0) = \frac{\pi}{4}.$$

(a) Let $u = x - y$. Express $\mathrm{d}u/\,\mathrm{d}x$ in terms of $\mathrm{d}y/\,\mathrm{d}x$.

(b) Show that

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sin^2 u.$$

(c) Solve the differential equation

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sin^2 u.$$

(d) Show that

$$y = x + \operatorname{arccot}(x + C),$$

where C is a constant.

(e) Determine the constant C such that $y(0) = \pi/4$.

Problem 269. (2022 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$(x^3 + y^3)\,\mathrm{d}x + xy^2\,\mathrm{d}y = 0.$$

(a) Let $y = ux$. Rewrite the given differential equation in terms of x and u.

(b) Find $A(x)$ and $B(u)$ such that

$$A(x)\,\mathrm{d}x + B(u)\,\mathrm{d}u = 0.$$

(c) Solve the differential equation

$$A(x)\,\mathrm{d}x + B(u)\,\mathrm{d}u = 0.$$

(d) Express the solution of the given differential equation in the form

$$g(x, y) = C,$$

where C is a constant.

Problem 270. (2022 Spring KAIST Differential Equations and Applications). Show that the functions

$$y_1(x) = x, \qquad y_2(x) = x \ln x, \qquad y_3(x) = x(\ln x)^2$$

are linearly independent on $(0, \infty)$, where ln denotes the natural logarithm.

Problem 271. (2022 Spring KAIST Differential Equations and Applications). Consider the second-order differential equation

$$y'' + y' - 2y = e^x \sin x + 2x - 1.$$

(a) Find the complementary solutions of

$$y'' + y' - 2y = 0.$$

(b) Find a particular solution of

$$y'' + y' - 2y = e^x \sin x.$$

(c) Find a particular solution of

$$y'' + y' - 2y = 2x - 1.$$

(d) Find a particular solution of

$$y'' + y' - 2y = e^x \sin x + 2x - 1.$$

(e) Find the general solution of

$$y'' + y' - 2y = e^x \sin x + 2x - 1.$$

Problem 272. (2022 Spring KAIST Differential Equations and Applications). We will use the Green's function to solve the initial-value problem

$$y'' + y' + y = e^{-x/2}, \qquad y(0) = 0, \qquad y'(0) = 1.$$

(a) Find the auxiliary equation for the homogeneous equation

$$y'' + y' + y = 0.$$

(b) Find two independent solutions $y_1(x)$ and $y_2(x)$ of the homogeneous equation.

(c) Compute the Wronskian of $y_1(x)$ and $y_2(x)$.

(d) Find the Green's function $G(x, t)$.

(e) Using the Green's function, find a particular solution $y_p(x)$ of

$$y'' + y' + y = e^{-x/2}$$

that satisfies $y_p(0) = y_p'(0) = 0$.

(f) Find a solution $y_h(x)$ of

$$y'' + y' + y = 0, \qquad y(0) = 0, \qquad y'(0) = 1.$$

(g) Find a solution of

$$y'' + y' + y = e^{-x/2}, \qquad y(0) = 0, \qquad y'(0) = 1.$$

Problem 273. (2022 Spring KAIST Differential Equations and Applications). Consider the Laplace transform $\mathcal{L}\{f(t)\}(s)$ of functions $f(t)$, $0 \le t < \infty$.

(a) Compute the Laplace transform of $f(t) = e^t$ using the definition of the Laplace transform.

(b) Assume that (i) $f(t)$ is differentiable, (ii) $\mathcal{L}\{f(t)\}(s)$ exists, and (iii) $\mathcal{L}\{f'(t)\}(s)$ exists. Prove that

$$\mathcal{L}\{f'(t)\}(s) = -f(0) + s\mathcal{L}\{f(t)\}(s).$$

(c) Take $f(t) = e^t$ in part (b), and find the Laplace transform of $f(t) = e^t$. *(Hint. The answer should coincide with the result from part (a).)*

(d) Solve the differential equation

$$y''(t) + y(t) = 0, \qquad y(0) = a_0, \qquad y'(0) = a_1,$$

on $(0, \infty)$ by using the Laplace transform.

Problem 274. (2022 Spring KAIST Differential Equations and Applications). For each $t \geq 0$, let

$$f(t) = \int_0^\infty \frac{\sin(tx)}{x}\, \mathrm{d}x$$

and

$$I(x) = \int_0^\infty \sin(tx)e^{-st}\, \mathrm{d}t.$$

(a) Show that for $s > 0$,

$$\int_0^\infty f(t)e^{-st}\, \mathrm{d}t = \int_0^\infty \frac{1}{x} I(x)\, \mathrm{d}x.$$

(b) Show that

$$I(x) = \frac{1}{x} - \frac{s^2}{x^2} I(x)$$

for $s > 0$ and $x > 0$.

(c) Show that

$$I(x) = \frac{x}{x^2 + s^2}.$$

(d) Show that

$$\mathcal{L}\{f(t)\}(s) = \frac{\pi}{2s}.$$

(e) Show that

$$f(t) = \frac{\pi}{2}$$

for all $t > 0$.

Problem 275. (2022 Spring KAIST Differential Equations and Applications). Let $y(t)$ be a solution of the differential equation

$$ty'' + y' + ty = 0,$$

or equivalently

$$(ty')' + ty = 0,$$

such that $y(0) = 1$. It is called the Bessel function of the first kind of order $\nu = 0$. Let

$$h(t) = ty'(t), \qquad F(s) = \mathcal{L}\{y(t)\}(s), \qquad G(s) = \mathcal{L}\{y'(t)\}(s).$$

(a) Show that

$$s\mathcal{L}\{h(t)\}(s) - h(0) - F'(s) = 0.$$

(b) Show that

$$-sG'(s) - F'(s) = 0.$$

(c) Show that

$$G'(s) = F(s) + sF'(s).$$

(d) Show that

$$\frac{F'(s)}{F(s)} = -\frac{s}{s^2 + 1}.$$

(e) Show that

$$F(s) = \frac{C}{\sqrt{s^2 + 1}}.$$

(f) Determine C, and find $\mathcal{L}\{y(t)\}(s)$.

Problem 276. (2022 Spring KAIST Differential Equations and Applications). Let

$$y = \sum_{n=0}^{\infty} a_n x^n$$

be a nonzero series solution of the differential equation

$$y'' - (x^2 - 1)y = 0$$

about the point $x = 0$.

(a) Find the constants p and q such that for $n = 2, 3, 4, \ldots$,

$$(pn + q)((n + 2)a_{n+2} + a_n) = na_n + a_{n-2}.$$

(b) Find the series solution satisfying

$$y(0) = 1, \qquad y'(0) = 0.$$

Problem 277. (2022 Spring KAIST Differential Equations and Applications). Let

$$\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 1 & 0 \end{pmatrix}.$$

(a) Solve the homogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

(b) Solve the nonhomogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \begin{pmatrix} -\sin t \\ \sin t \end{pmatrix}.$$

Problem 278. (2022 Spring KAIST Differential Equations and Applications). Diagonalize the matrix

$$\mathbf{A} = \begin{pmatrix} -3 & -1 & -5 \\ -2 & 1 & -2 \\ 2 & 1 & 4 \end{pmatrix}.$$

(a) Solve the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

(b) Compute $e^{t\mathbf{A}}$. An answer as a product of three matrices is acceptable.

Problem 279. (2022 Spring KAIST Differential Equations and Applications). Let

$$\mathbf{A} = \begin{pmatrix} 5 & -6 \\ -2 & 1 \end{pmatrix}.$$

(a) Find a fundamental matrix $\mathbf{\Phi}(t)$ of the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

(b) Find a particular solution of the nonhomogeneous system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \begin{pmatrix} 4 \\ 11 \end{pmatrix}.$$

(c) Solve the initial-value problem

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \begin{pmatrix} 4 \\ 11 \end{pmatrix}, \qquad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Problem 280. (2022 Spring KAIST Differential Equations and Applications). Find two linearly independent series solutions, up to the first four nonzero terms, for the differential equation

$$2xy'' + 3y' - xy = 0.$$

You can leave products of integers without multiplying them. For example, the expression

$$\frac{1}{1 \cdot 3 \cdot 5 \cdot 7}$$

is good enough.

Problem 281. (2022 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = x(1 - x - y), \\ y' = y\left(\frac{3}{4} - y - \frac{x}{2}\right). \end{cases}$$

(a) Find the critical points.

(b) Find the Jacobian matrix at each critical point.

(c) Classify the critical points.

Problem 282. (2022 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = 2x - y, \\ y' = 5x - 2y. \end{cases}$$

(a) Find all the critical points of the system.

(b) Show that $(0, 0)$ is a center.

(c) Find

$$\left(\frac{\mathrm{d}x}{\mathrm{d}t}, \frac{\mathrm{d}y}{\mathrm{d}t}\right)$$

at the points $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.

(d) As t increases, does the point $(x(t), y(t))$ move around $(0, 0)$ in the clockwise or counterclockwise orientation?

Problem 283. (2022 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = -x + 2y - 4x^3, \\ y' = x - 4y. \end{cases}$$

(a) Find and classify the critical points of the system.

(b) Explain why the given system does not have a periodic solution. *(Hint. If there were a periodic solution, then consider a closed curve C and apply Green's theorem. Or, you may use the theorem given in the textbook.)*

(c) Show that for any constant $a > 0$, the region

$$D = \{(x, y) \mid x^2 + y^2 < a\}$$

is an invariant region. *(Hint. Take a point (x, y) on the boundary of D. Then $(-x, -y)$ has the same direction as the normal vector pointing inside D.)*

(d) If $(x(t), y(t))$ is a solution satisfying the initial condition

$$(x(0), y(0)) = (x_0, y_0),$$

then what is the long-term behavior of $(x(t), y(t))$ as $t \to +\infty$?

Problem 284. (2022 Fall KAIST Differential Equations and Applications). For each $x > 0$, let

$$f(x) = \begin{cases} \cos x \sin x, & x \le \pi, \\ -2 \sin x, & x > \pi. \end{cases}$$

Solve the differential equation

$$y' \cos x + y \sin x = f(x).$$

Find the solution satisfying $y(4\pi/3) = 0$ defined on the largest interval.

Problem 285. (2022 Fall KAIST Differential Equations and Applications). Find the one-parameter family of solutions of the differential equation

$$(y - x)y' + y = 1.$$

Determine whether it is the general solution. Find the solution satisfying $y(1) = 0$ defined on the largest interval.

Problem 286. (2022 Fall KAIST Differential Equations and Applications). Solve the differential equation

$$(1 + 2e^{-y^2}x^2)xyy' = \frac{1}{2}.$$

Find the solution satisfying

$$y(-2) = -\sqrt{\ln 4}$$

defined on the largest interval.

Problem 287. (2022 Fall KAIST Differential Equations and Applications). Find a function of the form $\phi(x) = ax^2 + bx + c$ satisfying the differential equation

$$(x^2 - 1)x^2y'' - (x^2 + 1)xy' + (x^2 + 1)y = 0.$$

Find the general solution of the equation. Find all solutions satisfying the boundary conditions

$$y(-1) = 2, \qquad y(1) = -2,$$

if any.

Problem 288. (2022 Fall KAIST Differential Equations and Applications). Find an integer n such that the function $\phi(x) = x^n$ satisfies the differential equation

$$x(x - 1)y'' + (3x - 2)y' + y = 0.$$

Find the general solution of the equation. Find the solution satisfying

$$y(1/2) = 0, \qquad y'(1/2) = 4.$$

Find all solutions satisfying

$$y(1/2) = \ln 4, \qquad y(2) = 0,$$

if any.

Problem 289. (2022 Fall KAIST Differential Equations and Applications). Use the Green's function to solve the initial-value problem

$$y'' - 6y' + 9y = e^{4x}, \qquad y(0) = 1, \qquad y'(0) = -1.$$

Problem 290. (2022 Fall KAIST Differential Equations and Applications). Find the general solution to the system

$$\begin{cases} x' = y, \\ y' = z, \\ z' = w, \\ w' = x, \end{cases}$$

where x, y, z, w are all functions of t.

Problem 291. (2022 Fall KAIST Differential Equations and Applications). Solve the Cauchy–Euler equation

$$x^3y''' + 6x^2y'' + 7xy' + y = 0$$

on $(0, \infty)$.

Problem 292. (2022 Fall KAIST Differential Equations and Applications). Solve

$$y'''(y'')^{-1/2}e^{-x} = 1$$

with initial conditions

$$y(0) = y'(0) = 0, \qquad y''(0) = 1.$$

Problem 293. (2022 Fall KAIST Differential Equations and Applications). Solve the initial-value problem

$$y''' - y = 1, \qquad y(0) = y'(0) = y''(0) = 0$$

with Laplace transform.

Problem 294. (2022 Fall KAIST Differential Equations and Applications). Let $q > 0$ be a constant. For each $t \geq 0$, let

$$f(t) = \begin{cases} \dfrac{2t}{q}, & 0 \leq t \leq \dfrac{q}{2}, \\ 2\left(1 - \dfrac{t}{q}\right), & \dfrac{q}{2} \leq t < q, \end{cases}$$

and $f(t + q) = f(t)$. Show that

$$\mathcal{L}\{f(t)\}(s) = \frac{2}{qs^2} \tanh\left(\frac{qs}{4}\right).$$

Problem 295. (2022 Fall KAIST Differential Equations and Applications). Solve the initial-value problem

$$xy'' - 2(1 - x)y' - 2(1 - x)y = 2e^{-x}\cos x, \qquad y(0) = 0, \qquad y'(0) = -1.$$

Problem 296. (2022 Fall KAIST Differential Equations and Applications). Find the series solutions of the differential equation

$$(1 - x^2)y'' - 2xy' + 6y = 0$$

about the point $x = 0$.

Problem 297. (2022 Fall KAIST Differential Equations and Applications). Let $\nu > 0$ be a constant. Let $\lambda = 1$ or -1. Find the series solutions of the differential equation

$$x^2y'' - 2xy' + (2 - \lambda\nu^2x^2)y = 0$$

about the point $x = 0$. Identify the elementary functions represented by the series.

Problem 298. (2022 Fall KAIST Differential Equations and Applications). Find the series solutions of the differential equation

$$-x(1+x)^2y'' + (x^2-1)y' + (1-x)y = 0$$

about the point $x = -1$. Identify the elementary functions represented by the series.

Problem 299. (2022 Fall KAIST Differential Equations and Applications). Consider the linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 2 & 1 & 0 \end{pmatrix}.$$

(a) Find the eigenvalues of $\mathbf{A}$.
(b) Find a matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is diagonal.
(c) Let $\mathbf{x} = \mathbf{P}\mathbf{y}$. Construct a system of differential equations for $\mathbf{y}$.
(d) Solve $\mathbf{y}$.
(e) Solve $\mathbf{x}$.

Problem 300. (2022 Fall KAIST Differential Equations and Applications). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Find $\mathbf{A}^k$, where k is an arbitrary positive integer.
(b) Find $e^{t\mathbf{A}}$, where t is an arbitrary real number.
(c) Solve the linear system $\mathbf{x}' = \mathbf{A}\mathbf{x}$.

Problem 301. (2022 Fall KAIST Differential Equations and Applications). A bead with mass m slides along a curve

$$z = x^n e^{-x}.$$

Here n is a positive constant and x takes values on $(0, \infty)$. The damping force is $-\beta v$, where $\beta > 0$ is a constant and v is the velocity of the bead.

(a) Write down the system of differential equations describing the model.
(b) Find and classify the critical points of the system.

Problem 302. (2022 Fall KAIST Differential Equations and Applications). Write

$$x'' = \ln\left(\frac{1}{2} + x^2 + 2xx' + (x')^2\right)$$

as a system of differential equations. Then find and classify the critical points of the system.

Problem 303. (2022 Fall KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = -x^2 + 2y^2, \\ y' = -y^3 + y. \end{cases}$$

(a) Find the critical point (x_0, y_0) that is in the first quadrant.

(b) Show that if a constant $c > 0$ is small enough, then

$$\left\{(x, y) \mid (x - x_0)^2 + (y - y_0)^2 < c^2\right\}$$

is an invariant region. *(Hint. The change of variables $u = x - x_0$, $v = y - y_0$ may make the computation easier.)*

Problem 304. (2023 Spring KAIST Differential Equations and Applications). Determine whether each of the following statements is true or false.

(1) Every separable first-order equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$$

is exact.

(2) The substitution $u = y - 2x + 3$ makes the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 + \sqrt{y - 2x + 3}$$

separable.

(3) A population of bacteria growing exponentially doubles in population every 10 days. Therefore, the population size is described by the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{5}y,$$

where $t = 1$ corresponds to 1 day.

(4) A model for the population $P(t)$ of the fishery at time t is given by

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -(P - 400)(P - 1).$$

With this model, the fishery population becomes extinct.

(5) The initial-value problem

$$xy'' - y' = 0, \qquad y(0) = 0, \qquad y'(0) = 1$$

has a unique solution.

Problem 305. (2023 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$y' + y = x.$$

(a) First, solve the differential equation $y' + y = 0$.

(b) Find a particular solution of the form

$$y_p(x) = Ax + B$$

satisfying $y_p' + y_p = x$, where A and B are constants.

(c) Find the general solution of $y' + y = x$.

(d) Find a solution of $y' + y = x$ satisfying $y(0) = 2$.

Problem 306. (2023 Spring KAIST Differential Equations and Applications). Consider a solution $y(x)$ of the differential equation

$$y' = y^2 - 1.$$

(a) *(True/False)* If the solution y satisfies $y(0) = \frac{1}{2}$, then y cannot be a constant function.
(b) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = -\frac{3}{2}$. You do not need to find the formula for $y(x)$.
(c) *(True/False)* For the solution y in part (b), $\lim_{x\to\infty} y(x)$ exists but is not -1.
(d) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = 0$. You do not need to find the explicit formula for $y(x)$.
(e) For the solution y in part (d), show by direct computation that

$$\lim_{x\to\infty} y(x) = -1.$$

Problem 307. (2023 Spring KAIST Differential Equations and Applications). Consider a solution $y(x)$ of the differential equation

$$y' = e^{-y}.$$

(a) Find the solutions $y(x)$ satisfying $y(0) = -1$, $y(0) = 0$, and $y(0) = 1$.
(b) Sketch the solutions $y(x)$ satisfying $y(0) = -1$, $y(0) = 0$, and $y(0) = 1$.
(c) Define $z(x) = e^{-y(x)}$. Show that

$$z' = -z^2.$$

Problem 308. (2023 Spring KAIST Differential Equations and Applications). A tank contains 1000 liters of a solution consisting of 100 kg of salt dissolved in water. Pure water is pumped into the tank at the rate of 5 liters per second, and the mixture, kept uniform by stirring, is pumped out at the same rate. How long will it take until only 10 kg of salt remains in the tank?

Problem 309. (2023 Spring KAIST Differential Equations and Applications).

(a) *(True/False)* $\ln x$ and $\ln(2x)$ are both solutions of

$$y'' + \frac{1}{x}y' = 0$$

on $(0, \infty)$.
(b) *(True/False)* $\ln x$ and $\ln(2x)$ are linearly independent on $(0, \infty)$.
(c) For $k \in \{4, 5, 6, \dots\}$, compute the Wronskian of

$$\ln x,\ \ln(2x),\ \dots,\ \ln(kx)$$

on $(0, \infty)$.
(d) For $k \in \{4, 5, 6, \dots\}$, determine whether

$$\ln x,\ \ln(2x),\ \dots,\ \ln(kx)$$

are linearly independent on $(0, \infty)$. Justify your claim.
(e) *(True/False)* $\ln x$, $\ln(2x)$, and $\ln(3x)$ are all solutions of

$$y''' + \frac{2}{x}y'' = 0$$

on $(0, 1)$.

(f) *(True/False)* $\ln x$, $\ln(2x)$, and $\ln(3x)$ are linearly independent on $(0, \infty)$.

(g) *(True/False)* If $k \in \{4, 5, 6, \dots\}$, then $\ln x$, $\ln(2x)$, $\dots$, $\ln(kx)$ are all solutions of

$$y^{(k)} + \frac{k-1}{x} y^{(k-1)} = 0$$

on $(0, \infty)$.

Problem 310. (2023 Spring KAIST Differential Equations and Applications). Find the general solution of the differential equation

$$y^{(4)} - 4y^{(3)} + 8y'' - 8y' + 4y = e^{-x}(5x + 4\sin x).$$

Problem 311. (2023 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$x^2 y'' - 2xy' + x^2 y = x^4. \tag{1}$$

(a) Let

$$T(y) = x^2 y'' - 2xy' + x^2 y.$$

Calculate $T(\cos x)$, $T(\sin x)$, $T(x\cos x)$, and $T(x \sin x)$.

(b) Find a fundamental set of solutions of the associated homogeneous equation

$$T(y) = 0.$$

You may use the property

$$T(ay_1 + by_2 + cy_3 + dy_4) = aT(y_1) + bT(y_2) + cT(y_3) + dT(y_4)$$

for real numbers a, b, c, d and functions y_1, y_2, y_3, y_4.

(c) Find the general solution of (1).

Problem 312. (2023 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$x^2 y'' + 5xy' + 5y = 0, \qquad x > 0.$$

(a) *(True/False)* There is a real number r such that x^r is a solution of the equation.

(b) Let a be a positive constant. Find the solution of the equation subject to

$$y(a) = -a^2, \qquad y'(a) = 2a.$$

Problem 313. (2023 Spring KAIST Differential Equations and Applications). Consider

$$y'' + 2y' + y = x$$

with conditions

$$y(0) = 0 \tag{2}$$

and

$$y(1) = 0. \tag{3}$$

(a) *(True/False)* There is a nonconstant solution of the corresponding homogeneous equation satisfying conditions (2) and (3).

(b) Find the general solution of the corresponding homogeneous equation.

(c) Find two linearly independent solutions of the corresponding homogeneous equation such that one of them satisfies (2) and the other satisfies (3).
(d) Find the Green's function $G(x,t)$ of the two solutions you found.
(e) *(True/False)*

$$\phi(x) = \int_0^1 G(x,t)t \, \mathrm{d}t$$

is a particular solution of the above differential equation but does not satisfy $\phi(0) = \phi(1)$.
(f) Find a solution of the original nonhomogeneous equation subject to conditions (2) and (3).

Problem 314. (2023 Spring KAIST Differential Equations and Applications). Set

$$f(t) = \begin{cases} \sin t, & t < \dfrac{\pi}{2}, \\ \cos t, & t \geq \dfrac{\pi}{2}. \end{cases}$$

(a) Write $f(t)$ in terms of the Heaviside function, i.e., the unit step function.
(b) Find the Laplace transform of $f(t)$.
(c) Solve

$$y''(t) - y(t) = f(t)$$

subject to $y(0) = 0$ and $y'(0) = 1$.

Problem 315. (2023 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = y + t^2, \\ y' = z + t, \\ z' = x, \end{cases}$$

where x, y, and z are all functions of t.

(a) Use the method of elimination to obtain a differential equation involving only x and t.
(b) Find $x(t)$.
(c) Use the expression for $x(t)$ and the first equation in the system to find $y(t)$.
(d) Use the expression for $y(t)$ and the second equation in the system to find $z(t)$.

Problem 316. (2023 Spring KAIST Differential Equations and Applications). Determine whether each of the following statements is true or false.

(1) Since $x = 0$ is an irregular singular point of

$$x^3y'' - xy' + y = 0,$$

the differential equation possesses no solution that is analytic at $x = 0$.
(2) The general solution of

$$x^2y'' + xy' + (x^2 - 1)y = 0$$

is

$$y = c_1J_1(x) + c_2J_{-1}(x),$$

where $J_1(x)$ and $J_{-1}(x)$ are Bessel functions of the first kind of orders 1 and -1, respectively.
(3) The Bessel function of the second kind $Y_\nu(x)$ is linearly independent of $J_{-\nu}(x)$ for noninteger ν.
(4) The superposition principle for solutions of a system of linear equations holds only for homogeneous cases.

(5) The general solution of the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

with a real symmetric $n \times n$ matrix $\mathbf{A}$ is always written using the eigenvalues and eigenvectors of $\mathbf{A}$.

Problem 317. (2023 Spring KAIST Differential Equations and Applications). Let a and b be two nonzero real constants. For $t \geq 0$, let

$$f_a(t) = e^{at} \qquad \text{and} \qquad f_b(t) = e^{bt}.$$

(a) Assume that $a = b$. Compute $(f_a * f_a)(t)$ for $t \geq 0$.

(b) Assume that $a \neq b$. Compute $(f_a * f_b)(t)$ for $t \geq 0$.

(c) *(True/False)* For all a, b, there exist $M > 0$ and $c > 0$ such that

$$\left|(f_a * f_b)(t)\right| \leq Me^{ct}$$

for all $t \geq 0$.

(d) Prove that

$$\lim_{b \to a} (f_a * f_b)(t) = (f_a * f_a)(t)$$

for all $t \geq 0$.

(e) *(True/False)* For $b \neq a$,

$$(\cos bt * \cos at)(t) = \frac{\sin bt - \sin at}{b^2 - a^2}$$

for all $t \geq 0$.

Problem 318. (2023 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$y''(t) + 2y'(t) + 5y(t) = 0$$

with $y(0) = 2$ and $y'(0) = 3$. In the following, solve the differential equation with the given initial conditions using the Laplace transform.

(a) Apply the Laplace transform $\mathcal{L}$ to the above differential equation and obtain

$$p(s)\mathcal{L}\{y(t)\}(s) - q(s) = 0,$$

where $p(s)$ and $q(s)$ are polynomials in s. Find $p(s)$ and $q(s)$.

(b) Find the function $h(t)$ whose Laplace transform is

$$\frac{q(s)}{p(s)}.$$

(c) Check that $h(t)$ is a solution of the above differential equation.

(d) *(True/False)* For a Riemann-integrable function $f(t)$,

$$y(t) = \frac{1}{2}\int_0^t e^{-\tau} \sin(2\tau) f(t-\tau)\, \mathrm{d}\tau$$

is a solution of

$$y''(t) + 2y'(t) + 5y(t) = f(t), \qquad y(0) = y'(0) = 0, \qquad t \geq 0.$$

Problem 319. (2023 Spring KAIST Differential Equations and Applications). Consider the differential equation

$$xy'' + 2y' + xy = 0, \qquad x > 0. \tag{1}$$

(a) Classify the singular points, if any.

(b) Let

$$T(y) = xy'' + 2y' + xy.$$

Compute $T(x^{n+r})$ for a natural number n and a real number r.

(c) Find the conditions on c_n and r such that

$$\sum_{n=0}^{\infty} c_n x^{n+r}$$

is a solution of equation (1), i.e.,

$$T\left(\sum_{n=0}^{\infty} c_n x^{n+r}\right) = \sum_{n=0}^{\infty} c_n T(x^{n+r}) = 0.$$

(d) Find the series solutions of equation (1) and show that they form a fundamental set of solutions of equation (1).

(e) Find the smallest $l > 0$ such that

$$-\infty < \lim_{x \to 0^+} x^l y(x) < \infty$$

for every y that satisfies equation (1), and prove it.

Problem 320. (2023 Spring KAIST Differential Equations and Applications). Find the solution of the system

$$\mathbf{x}' = \begin{pmatrix} 0 & -1 \\ 13 & 4 \end{pmatrix} \mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix} 0 \\ 3 \end{pmatrix}. \tag{2}$$

Let

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 13 & 4 \end{pmatrix}.$$

(a) *(True/False)* $\mathbf{A}$ has a real eigenvalue.

(b) Find the matrix exponential $e^{t\mathbf{A}}$ by using the Laplace transform and show that

$$e^{(t+s)\mathbf{A}} = e^{t\mathbf{A}} e^{s\mathbf{A}}$$

for all $t, s > 0$.

(c) Find the solution of system (2).

(d) *(True/False)* Let $\mathbf{x}(t)$ be the solution of (2). Suppose $\mathbf{y}(t)$ is the solution of

$$\mathbf{y}' = \begin{pmatrix} 0 & -1 \\ 13 & 4 \end{pmatrix} \mathbf{y}, \qquad \mathbf{y}(0) = e^{\pi} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Then there exists $t > 0$ such that

$$\mathbf{y}(t) \neq \mathbf{x}\left(t + \frac{\pi}{2}\right).$$

Problem 321. (2023 Spring KAIST Differential Equations and Applications). Consider the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{pmatrix} 1 & -3 & 3 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix}.$$

(a) Find the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\mathbf{A}$.

(b) Find a set of linearly independent vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ satisfying

$$\mathbf{A}\mathbf{v}_i = \lambda_i \mathbf{v}_i, \qquad \|\mathbf{v}_i\| \neq 0, \qquad i = 1, 2, 3. \tag{3}$$

(c) *(True/False)* Suppose $\{\mathbf{v}_i\}_{i=1}^3$ satisfies (3). Let

$$\mathbf{x} = \sum_{i=1}^{3} x_i \mathbf{v}_i.$$

Then $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is equivalent to

$$x_i'(t) = \lambda_i x_i(t), \qquad i = 1, 2, 3.$$

(d) Find the solution to the system satisfying the initial condition

$$\mathbf{x}(0) = \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}.$$

Problem 322. (2023 Spring KAIST Differential Equations and Applications). Consider the linear system

$$\mathbf{x}' = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{-2t} \sec^3 t.$$

Let

$$\mathbf{A} = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix}.$$

(a) Compute $e^{t\mathbf{A}}$.

(b) *(True/False)* $e^{t\mathbf{A}}$ is invertible for all $t \geq 0$.

(c) *(True/False)*

$$\mathbf{u}(t) = \int_0^t e^{-s\mathbf{A}} \begin{pmatrix} 2 \\ 0 \end{pmatrix} (e^{-2s} \sec^3 s) \, \mathrm{d}s$$

is a solution of the above linear system for $t \geq 0$.

(d) Find the general solution of the above linear system.

Problem 323. (2023 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = x - 2y, \\ y' = 2x - y. \end{cases}$$

(a) Classify the critical point of this system.

(b) Find the velocity (x', y') at the point $(c, 0)$, where $(c, 0)$ is an arbitrary point on the positive half of the x-axis of the xy-plane.

(c) Draw the curve of a nontrivial solution $(x(t), y(t))$ on the xy-plane, with an arrow on it to show the direction of motion of the point $(x(t), y(t))$ as t increases.

Problem 324. (2023 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = -2x + 2x^2, \\ y' = -3x + y + 3x^2. \end{cases}$$

(a) Find the critical points of this system.
(b) Find the Jacobian matrix at each critical point.
(c) For each critical point, find the trace and determinant of the Jacobian matrix.
(d) Classify each critical point.

Problem 325. (2023 Spring KAIST Differential Equations and Applications). Consider the system

$$\begin{cases} x' = y, \\ y' = -x + y(1 - x^2 - y^2). \end{cases} \tag{4}$$

(a) Find the critical points of this system.
(b) Suppose $(x(t), y(t))$ is a solution of (4). Let

$$r(t) = x(t)^2 + y(t)^2.$$

Derive an equation satisfied by

$$r' = 2xx' + 2yy'.$$

(c) Suppose $r(0) > 1$. Prove that

$$r(t) < r(0)$$

for all $t > 0$ by using the equation derived in part (b).
(d) *(True/False)* The region

$$R = \{(x, y) \mid x^2 + y^2 < c\}$$

is an invariant region of Type I when $c > 1$.
(e) Suppose $c > 1$ is a constant. Show that there is at least one periodic solution in the region

$$\{(x, y) \mid x^2 + y^2 < c\}.$$